

Modal Logic Neural Networks

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Abstract

Neural Networks are indispensable to natural sciences and society. Their impact extends from applications in public health to workforce productivity. Here, we introduce Modal Logic Neural Networks (MLNNs) – an end-to-end differentiable logical neural network realisation of modal logic which evaluates a learnable truth function across possible-world semantics. This neural architecture handles para-consistency and inconsistency via a learnable world accessibility relation and valuation function. Because the modality is fixed by which frame axioms the relation satisfies rather than by the operator, one differentiable engine covers the epistemic, doxastic, deontic and temporal readings, with applications from verification of reactive and distributed systems to legal discourse and microeconomic utility models. In this paper, we introduce a model of differentiable Kripke semantics, and establish their soundness, convergence, and structural guarantees. We show four applications, in which the learned relation reads as a trust matrix, an operating-regime embedding with safety bounds, a temporal precedence order, and a recovered constraint graph.

Keywords: Modal Logic, Kripke Semantics, Neurosymbolic AI, Differentiable Reasoning, Constraint Satisfaction

1. Introduction

While modern neural networks, especially large language models, primarily rely on statistical plausibility to generate outputs, Neurosymbolic AI combines connectionist learning and symbolic reasoning to achieve more human-like reasoning abilities. This addition of differentiable logical constraints to neural architectures has been predominantly using classical logic (DeepProbLog [Manhaeve et al. \(2018\)](#), Semantic Loss [Xu et al. \(2018\)](#), Scallop [Li et al. \(2023\)](#)). However given the longstanding versatility of modal logics, a modal neural architecture would greatly extend human-like expressibility and allow for a generalisation of previous neurosymbolic architectures. Via modal logics, one can permit modalities to logical constraints (formulas) which can add critical value such as: the ability to express a temporal safety rule ($\Box(\text{red} \rightarrow \neg \text{move})$), a deontic permission ($\Box\phi_{\text{SAFE}}$), epistemic knowledge ($K_a\phi$) and a doxastic belief ($B_a\phi$). Ultimately, truth is preserved across a flexible Kripke structure with meaningful handling of para-consistency.

This paper introduces Modal Logic Neural Networks (MLNNs) ¹, a differentiable instantiation of Kripke semantics in which the worlds, the valuation, and the accessibility relation $A_\theta \in [0, 1]^{|W| \times |W|}$ are all part of one forward pass, with the valuation, the relation, or both learnable, and truth carried as a bounded interval $[L, U]$ rather than a scalar (Łukasiewicz operators relax $\min / \max$ to $\text{smin}_\tau / \text{smax}_\tau$; Figure 1, formalised in Section 3).

¹Code available at <https://github.com/sulcantonin/torchmodal> (pip install torchmodal).

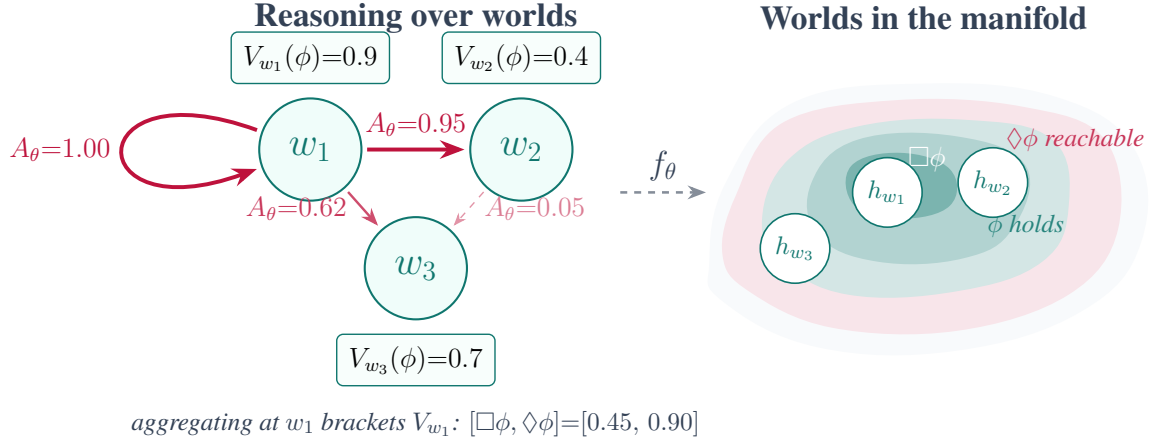


Figure 1: **Modal Logical Neural Networks.** **Left:** three worlds, each with its own truth value $V_w(\phi)$, joined by learned accessibility weights $A_\theta \in [0, 1]$ saying how much one world’s verdict bears on another’s. The weights are the parameters. One modal neuron firing at w_1 (Eq. 2) gives necessity $\Box\phi(w_1) = \text{smin}_\tau[(1 - A_\theta) + V] \approx 0.45$ (“how true in *every* world w_1 sees”) and possibility $\Diamond\phi(w_1) = \text{smax}_\tau[A_\theta + V - 1] \approx 0.90$ (“in *some*”). They bracket $V_{w_1}=0.9$, so the bound is consistent and $L_{\text{contra}}=0$; had they crossed, that gap is the training signal. **Right:** the encoder f_θ turns worlds into points h_w . The A_θ computing the answer is the A_θ the analyst reads.

Contributions. (1) A differentiable, end-to-end framework of a learnable world accessibility relation A_θ and valuation V_ϕ . The accessibility relation can be **T** reflexive, **4** transitive, **B** symmetric, **5** Euclidean, **D** serial (no dead ends). These properties are denoted as frame axioms, and are measurable after training so need not be imposed on the Kripke structure. (2) Interpretability is built into inference with two modes deductive and inductive. To showcase, four real world examples are presented, where the final outputs are domain-specific and human-interpretable. (3) An open-source pythonic framework, called *torchmodal*, based on PyTorch¹ an ubiquitous open-source Meta library. *torchmodal* ships modal logic operators, accessibility relation parameterisations, and the frame-axiom audit, together with the four examples. (4) A classical logic neural network as the zero-one special case of MLNNs, making MLNNs a generalisation encompassing classical LNNs.

2. Related Work

Neurosymbolic AI. DeepProbLog [Manhaeve et al. \(2018\)](#) combines neural networks with probabilistic logic programming; the Semantic Loss [Xu et al. \(2018\)](#) enforces propositional constraints on neural outputs, functionally close to our L_{contra} but single-world; Scallop [Li et al. \(2023\)](#) and NeurASP [Yang et al. \(2020\)](#) give differentiable probabilistic Datalog and ASP; Logic Tensor Networks [Badreddine et al. \(2022\)](#) ground first-order logic in real-valued semantics.

Modal Logic in AI. Modal logic has a long history in AI for reasoning about knowledge, belief, time, and obligation [Fagin et al. \(1995\)](#); [Hintikka \(1962\)](#); [Liau \(2003\)](#), and supplies the logical foundations of multiagent systems [Shoham and Leyton-Brown \(2009\)](#). Symbolic tools such as ml-solver [Rohkohl \(2024\)](#) build Kripke structures and check axiom sets for contradictions or tautologies before deployment; MLNN is complementary, fitting the accessibility relation to data and exposing it as the inspectable forward-pass artefact rather than requiring it to be specified in advance.

Fuzzy modal logic and smooth min/max. Many-valued and Łukasiewicz modal logics with fuzzy accessibility are established [Fitting \(1991\)](#); [Bou et al. \(2011\)](#); [Hájek \(2001\)](#); [Caicedo and Rodríguez](#)

(2010) and are essentially the semantics our forward pass computes; what is new is that the frame is a *learned parameter* rather than a fixed structure.

Logical and Modal Neural Networks. LNNs Riegel et al. (2020) introduced weighted real-valued logic with provable bounds for the propositional case; we extend the re-grounding to modal logic by making the Kripke triple (W, R, V) an explicit, either-fixed-or-learnable component, with $\Box/\Diamond$ realised as bounded sound aggregators rather than recurrent transition operators. Connectionist Modal Logic Garcez et al. (2007) embeds modal operators in recurrent hidden states and, like MLNN, is trained end-to-end, but its accessibility relation is entangled in those hidden states and never materialised; our A_θ is an explicit, inspectable first-class object. STLCG Leung et al. (2023) addresses Signal Temporal Logic via robustness values; our $[L, U]$ formulation separates epistemic uncertainty from degree of satisfaction and generalises beyond fixed temporal ordering. Recent work on temporal-logic satisfiability Luo et al. (2022) and epistemic logic in RL Engesser et al. (2025) addresses specific modal fragments; in contrast, MLNN provides a single, end-to-end-differentiable framework in which the accessibility relation itself is learned by gradient descent. The Łukasiewicz choice for Boolean connectives follows van Krieken et al. (2022); MLNs Richardson and Domingos (2006) and ProbLog De Raedt et al. (2007) combine first-order logic with probability but do not natively support modal operators. Differentiable constraint solvers such as SATNet Wang et al. (2019) (a learnable MaxSAT layer) and recurrent relational networks Palm et al. (2018) solve combinatorial problems by learning the constraint rules from a corpus of solved instances; MLNN instead solves each instance directly from explicit modal axioms and additionally learns the accessibility relation itself, a complementarity we make precise on graph colouring (Section 5.4). In several instantiations (C-MAPSS, ROCStories) this relation is metric-learned from an embedding f_θ , an additional interpretability layer atop the provable frame axioms; both the valuation V and the accessibility A_θ can be learned, differentiable, and inspected.

3. Method: Modal Logic Neural Networks

3.1. MLNN Object Language

The MLNN supports the following grammar of logical formulae:

$$\phi ::= p \mid \neg\phi \mid \phi \wedge \psi \mid \phi \vee \psi \mid \phi \rightarrow \psi \mid \Box\phi \mid \Diamond\phi \mid \phi \mathcal{U} \psi \quad (1)$$

where p ranges over atomic propositions. We define the set of well-formed formulas inductively over the formulas in 1 above. Boolean connectives use Łukasiewicz fuzzy logic van Krieken et al. (2022): $\neg\phi = 1 - \phi$; $\phi \wedge \psi = \max(0, \phi + \psi - 1)$; $\phi \vee \psi = \min(1, \phi + \psi)$; $\phi \rightarrow \psi = \min(1, 1 - \phi + \psi)$. Each formula ϕ is represented by truth bounds $[L_\phi, U_\phi] \subseteq [0, 1]$ per world, forming a DAG-structured formula graph where nodes are subformulae and edges are dependencies. Inference propagates bounds through this graph using the upward-downward bound-propagation algorithm of LNNs Riegel et al. (2020), detailed in the supplementary material (*Complete Theoretical Analysis*).

3.2. Kripke Semantics

A modal Kripke frame $F = (W, R)$ consists of a non-empty set (of worlds) W , and a binary world accessibility relation $R \subseteq W \times W$. A valuation V of the object language on a frame $F = (W, R)$ is a map associating with each proposition p a set of worlds $V(p)$. This is understood as the set of worlds at which p is true. A Kripke model is a pair $M = (F, V)$ where F is a frame and V is a valuation on F . We define a truth-relation $(M, x) \models \phi$ where ϕ is a well-formed formula of the object language to mean formula ϕ is true at world x in model M . A formula is true in a model $M = (F, V)$ if for every $w \in W$, $w \models \phi$. A formula ϕ is valid in a frame F if ϕ is true in all models

based on F , in symbols $F \models \phi$. We define modal logic K in the object language as the set of all formulas that are valid in all modal Kripke frames.

MLNN is a differentiable, learnable instance of this, where a “world” may be an agent’s belief state, a moment in time, or any context. Truth values become bounds $[L, U] \subseteq [0, 1]$, and the inter-world structure is either a fixed crisp R or a learnable $A_\theta \in [0, 1]^{|W| \times |W|}$, optionally sparsified by top- k masking to $\tilde{A}$ (supplement, *Masking and Sparsity*). The operators $\Box$ (necessity) and $\Diamond$ (possibility) can have any modality interpretation; epistemic (K_a), doxastic (B_a), temporal (G/F) or deontic (O/P). The interpretation is fixed by the user. With `torchmodal` only one computational engine is needed. The user specifies the frame axioms A_θ as needed. For an example, one can specify a reflexive and transitive frame and obtain $S4$ modal Kripke model. Temperature τ controls the soft aggregations; training minimises L_{task} and the contradiction loss L_{contra} , balanced by β .

3.3. Differentiable Kripke Semantics

Each Kripke component becomes a differentiable tensor. For proposition p the MLNN stores truth bounds of shape $(|W|, 2)$, each row $[L_{p,w}, U_{p,w}]$ giving the bounds of p in world w . A scalar valuation conflates two distinct uncertainties, how strongly evidence supports p and how strongly it rules p out; the pair separates them (L guarantees how true p is, U how not-false), with scalars the degenerate case $L=U$. Two reasons to carry both. (i) Many evidence sources emit two numbers: an NLI head returns separate entailment and contradiction probabilities, and setting $L=\text{entail}$, $U=1 - \text{contradict}$ lets an internally inconsistent source surface directly as $L > U$. (ii) The same shape expresses the bound contradiction $\max(0, L_\phi - U_\phi)$, an inconsistency no classical assignment satisfies.

3.3.1. MODAL OPERATORS: THE NECESSITY AND POSSIBILITY NEURONS

The modal operators are neurons that aggregate across worlds. Classical logic takes hard min/max over a fixed neighbourhood; we use differentiable relaxations over the weighted $\tilde{A}$ so gradients pass through the structural decision. For truth values $x = \{x_i\}$, $\text{smin}_\tau(x) = -\tau \log \sum_i \exp(-x_i/\tau)$ is a sound lower bound on $\min(x)$ and $\text{smax}_\tau(x) = \tau \log \sum_i \exp(x_i/\tau)$ a sound upper bound on $\max(x)$;² and $\text{conv-pool}_\tau(x, z) = \sum_i w_i x_i$ with $w_i = \text{softmax}(z_i/\tau)$ is a convex pool, lower-bounding $\max(x)$ at $z=x$ and upper-bounding $\min(x)$ at $z=-x$. Default $\tau = 0.1$.

The $\Box$ (Necessity) and $\Diamond$ (Possibility) Neurons $\Box\phi$ is a weighted universal quantification (“weakest link detector”) and $\Diamond\phi$ a weighted existential (“evidence scout”). With $\bar{A}_{w,w'} := 1 - \tilde{A}_{w,w'}$,

$$\begin{aligned} L_{\Box\phi,w} &= \text{smin}_{\tau}(\bar{A}_{w,w'} + L_{\phi,w'}), & U_{\Box\phi,w} &= \text{conv-pool}_{\tau}(\bar{A}_{w,w'} + U_{\phi,w'}), \\ L_{\Diamond\phi,w} &= \text{conv-pool}_{\tau}(\tilde{A}_{w,w'} + L_{\phi,w'} - 1), & U_{\Diamond\phi,w} &= \text{smax}_{\tau}(\tilde{A}_{w,w'} + U_{\phi,w'} - 1). \end{aligned} \quad (2)$$

Each conv-pool_τ above takes the bounding direction of its definition: $z=-x$ for $U_{\Box\phi}$ (an upper bound on min) and $z=x$ for $L_{\Diamond\phi}$ (a lower bound on max), with x the bracketed argument shown. The duality $\Diamond\phi \equiv \neg\Box\neg\phi$ holds via $\text{smax}(x) = 1 - \text{smin}(1 - x)$ (derived in the supplementary material, *De Morgan Duality of the Modal Operators*). All four bounds in Equation 2 are clamped to $[0, 1]$ before use: the explicit clamp matters because smin_τ can exceed 1 on a single-term bracket, which would break the $[0, 1]$ truth-bound invariant.

² smin/smax distinguish these log-sum-exp aggregators from the probability-normalising softmax.

The Until ($\mathcal{U}$) Neuron. When A_θ encodes a temporal order, the until $\phi \mathcal{U} \psi$ (“ ϕ holds until ψ becomes true”) is a third modal aggregation. Over worlds in time order $1, \dots, T$ its bounds follow the backward Łukasiewicz recurrence $(\phi \mathcal{U} \psi)_t = \psi_t \vee (\phi_t \wedge (\phi \mathcal{U} \psi)_{t+1})$, terminating at $(\phi \mathcal{U} \psi)_T = \psi_T$, applied to the L - and U -bounds in turn (equivalently $\bigvee_{t' \geq t} \psi_{t'} \wedge \bigwedge_{t \leq s < t'} \phi_s$); the sentence-ordering experiment (Section 5.3) scores a single such until at the first world.

Graceful Handling of Contradictions. The $[L, U]$ bound representation detects and localises contradictions in user-provided axioms rather than silently producing inconsistent outputs. Atomic propositions and non-modal compound nodes are constructed so that $L_{\phi, w} \leq U_{\phi, w}$ holds by construction. Modal compound nodes, however, can produce $L_{\phi, w} > U_{\phi, w}$ during the upward pass when the current accessibility relation A_θ is incompatible with the user-supplied axioms; this inversion is not projected back to a valid interval but surfaced as the per-formula, per-world residual $\max(0, L_{\phi, w} - U_{\phi, w})$ that contributes to L_{contra} . Gradient descent then resolves the inversion by adjusting the propositional content (deductive mode), the accessibility structure (inductive mode), or both. This contrasts with single-valued robustness approaches (e.g., STLCCG), where conflicting constraints can cancel silently.

3.4. Flexible and Learnable Accessibility Relations

The defining capability is that the accessibility relation can be a learnable parameter A_θ rather than the fixed R of classical modal logic. Both modes are exercised: colouring uses a fixed adjacency when solving a given instance, while C-MAPSS learns A_θ as a metric kernel and the inductive colouring task recovers it from data alone (Section 5). We parameterise $A_\theta : W \times W \rightarrow [0, 1]$ either as a matrix of learnable logits through a sigmoid (small domains) or, for scale, by metric learning: an encoder maps each world to $\mathbf{h}_w \in \mathbb{R}^d$ and $A(w_i, w_j) = \sigma(\mathbf{h}_{w_i}^\top \mathbf{h}_{w_j})$ makes logical access geometric proximity, reducing parameters from quadratic to linear in $|W|$. The weights enter the modal neurons through the implication $1 - (\tilde{A})_{ij} + L_{\phi, w_j}$ inside the smin of Equation 2: at $(\tilde{A})_{ij} \approx 1$ the truth value fully participates in the minimum, at ≈ 0 it is removed. Gradient descent on the contradiction loss updates θ , which is the inductive capability that lets the model discover the structure best resolving the data’s contradictions.

3.5. Loss Function and Training

Training minimises $L_{\text{total}} = L_{\text{task}} + \beta L_{\text{contra}}$ (Eq. 3) end to end, where L_{task} is any supervised loss and L_{contra} penalises crossed bounds (Eq. 3.5):

$$L_{\text{total}} = L_{\text{task}} + \beta L_{\text{contra}}, \quad L_{\text{contra}} = \sum_{w \in W} \sum_{\phi} \max(0, L_{\phi, w} - U_{\phi, w}). \quad (3)$$

β trades task accuracy against logical coherence. For pure constraint satisfaction we set $L_{\text{task}}=0$ and learn entirely from L_{contra} (*pure satisfiability mode*), making the MLNN a differentiable energy-based model whose free parameters are the proposal logits.

4. Theoretical Properties

The properties below are shown in full in the supplementary material under *Complete Theoretical Analysis*:

Soundness. An aggregator is *sound* when it never claims more than the operator it relaxes, so its output brackets that min / max and a satisfied axiom under the soft reading never gives false reassurance about the crisp one: $\text{smin}_\tau(x) \leq \min(x)$ and $\text{conv-pool}_\tau(x, -x) \geq \min(x)$ for $x_i \in [0, 1]$, with equality as $\tau \rightarrow 0$. This bounds the operators, not the graph: a modal node may still emit $L_\phi > U_\phi$, the contradiction signal described above.

Convergence. For acyclic graphs over the non-modal fragment, with polarity-tracked monotone operators and fixed parameters, each sweep is exact at $O(|V| + |E|)$ operator evaluations; *acyclicity* makes the count finite, the monotone-bounded argument alone giving only asymptotic convergence. The *joint* fixed point needs more than one round whenever axioms share a subformula, a downward update stales a sibling already consumed upward, so inference iterates to a tolerance. Cyclic graphs get no finite bound.

Structural recoverability. For each of T, 4, B, 5, D a differentiable regulariser $R_P \geq 0$ vanishes exactly on relations with that property, and adding it preserves soundness. Four qualifications. (i) For four of the five this is well-posedness, not a training-dynamics guarantee: R_P is the property’s residual, so its zero set is the property by construction; only seriality has content beyond the definition. (ii) No experiment here uses them and all four run in the audit regime, axioms measured post-hoc, so reported values are emergent and can fail silently, which is why we measure them. (iii) Exact satisfaction is unattainable under $A_\theta = \sigma(\tilde{A}_\theta)$, whose entries lie in the *open* $(0, 1)$: an exact 1.00 signals saturation or a kernel symmetry, not a crisp relation. (iv) The regularisers use the product t-norm and the audit Łukasiewicz; the former is strictly stronger, so $R_P=0$ implies the audit passes but not conversely.

Complexity. A modal neuron costs $O(|W|)$ under a fixed sparse relation and $O(|W|^2)$ dense; the metric parameterisation (Section 3.4) defines accessibility over d -dimensional embeddings ($d \ll |W|$), cutting parameters to $O(d \cdot |W|)$ and, with top- k masking, compute to $O(k \cdot |W|)$ per neuron. Forming that mask exactly still scores every pair, so end-to-end cost stays $O(|W|^2)$, as the supplement measures: the parameter saving removes the dense cliff, not the exponent.

5. Experiments

We evaluate MLNN on four examples spanning the modal range and both directions of use. The thesis is that *the object the model computes is the object the analyst reads*, delivered at accuracy competitive with strong task-specific baselines, which we report alongside; the contribution is the auditable modal structure those baselines do not express. One inference engine serves all four, every one of them an instance of Equation 3 under contradiction loss alone (C-MAPSS, colouring, ROCStories, doxastic). Each answers one question: whether a learned trust relation exposes an incoherent *stance*, and whether the modal layer adds anything its scorer does not (§5.1); whether one relation trained on a deontic contradiction supports several unrelated expert reads at once (§5.2); whether a modal operator is load-bearing or a propositional scorer on the same embeddings would do as well (§5.3); and, where ground truth for the relation exists, whether the learned relation is the correct one, i.e. whether the artefact is faithful rather than merely plausible (§5.4). Full baselines, per-seed tables and hyperparameters are in the appendices.

5.1. Doxastic Deception: a trust matrix that surfaces a saboteur the LLM-judge misclassifies

The *doxastic* reading $B_a\phi$ (“agent a believes ϕ ”) audits not what is true but whether an agent’s stance coheres: a multi-agent LLM system can be derailed by a member holding an incoherent belief without ever making a checkably false claim. Six agents, each the same locally-served LLM (via *ollama*), negotiate a \$50K budget over twelve rounds; agent 2 is a saboteur. A frozen NLI head scores every utterance against “the budget is fair” into a per-round (entail, contradict), summarised per agent as a bound $[L_j, U_j]$; a persistent flip-flopper drives $L_j > U_j$, a crossed bound no truth value holds. A multi-agent Kripke model trains the 6×6 trust matrix by maximising $\square_i$ coherent over the trust relation (the doxastic axiom $A_\theta[i, j] \rightarrow \text{consistent}(j)$, $\text{consistent}(j) = 1 - \text{relu}(L_j - U_j)$; Eq. 3.5), so a self-contradictory agent collapses its column toward zero and the predicted

saboteur is $\arg \min_j \overline{A_\theta[\cdot, j]}$. The baseline re-prompts the same LLM with the raw transcript to name the saboteur.

Result and interpretation. We run five backbones $\times$ three seeds per scenario, so every cell below is a detection count out of 15, and we report a Wilson 95% interval with it. Because MLNN and the judge read the *same* transcript, the two are paired and we test them with an exact McNemar test rather than a two-proportion test. Three regimes appear. When the saboteur flip-flops blatantly both tie (15/15, CI [0.80, 1.00]): the transcript is obvious. The separation is the subtle case, where the saboteur drifts politely while a stylistically loud but content-honest agent looks suspicious. The LLM-judge, reading surface form, fingers the loud-honest decoy and scores 3/15 ([0.07, 0.45]); MLNN, whose NLI front-end scores entailment rather than sentiment, discounts the tone and reaches 12/15 ([0.55, 0.93]; McNemar $p = 0.004$, 9 discordant pairs, all favouring MLNN). Under prompt *injection*, where the saboteur embeds a clause framing an honest agent, MLNN is a clean 15/15 versus 5/15 ($p = 0.002$): the injection text lives in the saboteur’s own transcript and cannot move the framed agent’s column, an immunity the modal pipeline has by construction rather than by training.

What the modal layer does and does not contribute. The pipeline stacks three components and the gap above could belong to any of them, so we peel them apart on identical transcripts (supplement, *Doxastic*). The frozen NLI head *alone*, with no bound, no trust matrix and no training, already reaches 15/13/15 against the full stack’s 15/12/15, a difference significant in no scenario (McNemar $p = 1.0$). We state this plainly: the *detection* gap over the LLM-judge is attributable to the NLI front-end, not the modal layer, and a judge given the same front-end would close it. What the modal layer contributes is the object behind the verdict, a per-cell trust matrix with the utterances that drained each column (Figure 3), and the structural injection-immunity above, which follows from per-agent column isolation and which no transcript-level scorer has. That claim is narrower than a detection win, and all three scenarios are author-designed, so these numbers measure detection of a constructed failure mode rather than its prevalence in the wild.

5.2. Turbofan Wear (C-MAPSS): one objective, three reads of one relation

C-MAPSS Saxena et al. (2008) is NASA’s standard run-to-failure benchmark for turbofan engines; FD002 runs each engine from healthy to failure under six unlabelled operating regimes, recording 21 sensors, 3 operating settings and a ground-truth Remaining Useful Life (RUL) per cycle. Each (engine, cycle) pair is a Kripke world, and both Kripke components are learned from the 24-dimensional sensor state alone: a metric head gives the within-engine accessibility $A_\theta(w_i, w_j) = \sigma(s f_\theta(s_i)^\top f_\theta(s_j) + b)$ with $f_\theta : \mathbb{R}^{24} \rightarrow \mathbb{R}^{16}$ (2642 parameters), and a second head the valuation $V_\phi(s) = \sigma(\text{MLP}(s))$ (1665 parameters), $V_{\text{SAFE}} = 1 - V_\phi$. With deontic predicates $\text{SAFE} = (\text{RUL} > 100)$ and $\text{FAIL-SOON} = (\text{RUL} < 30)$, the regulator reads the per-cycle bounds $L_{\Box(\text{SAFE})}$ and $U_{\Diamond(\text{FAIL-SOON})}$. The heads carry disjoint gradients: $\mathcal{L}_V = \text{BCE}(V_\phi(s), \mathbb{I}[\text{RUL} < 30])$ trains the valuation, while A_θ is trained by the deontic contradiction alone, with V detached,

$$\mathcal{L}_{\text{deon}} = \underbrace{(1 - U_{\Diamond(\text{FAIL-SOON})})^2}_{\text{RUL} < 30} + \underbrace{U_{\Diamond(\text{FAIL-SOON})}^2 + (1 - L_{\Box(\text{SAFE})})^2}_{\text{RUL} > 100}, \quad (4)$$

the at-risk middle carrying no term and being shaped only through A_θ . No locality kernel or other constructed target is fitted. RUL supervises V_ϕ during training but is never read at inference: the safety envelope is therefore not an unsupervised read, whereas the regime and degradation probes are, since no loss references either.

Result and interpretation. This is the deontic case, and the point is applicability, not accuracy. From one rule where safe must hold, fail-soon may, never both and the model learns a per-cycle

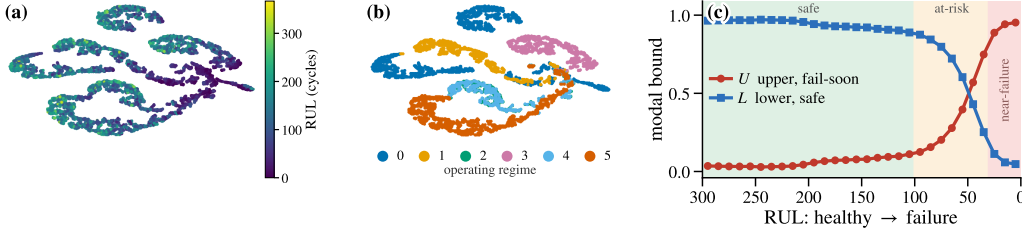


Figure 2: **Turbofan Wear: three reads of one relation trained only on the deontic contradiction.** (a) the embedding f_θ (t-SNE) coloured by RUL: worlds organise along a wear manifold. (b) the same embedding, same axes, coloured by operating regime. (c) the deontic bounds against RUL. Both V_ϕ and A_θ read sensors only; no RUL is read at inference. Panels are seed 45, quoted statistics over 3 seeds.

safety margin that tracks wear, extends to unlabelled operating states, and matches a standard classifier while staying inspectable. On the whole FD002 fleet (260 train /259 test engines, 48,299 worlds; 3 seeds, mean $\pm$ std), one contradiction objective supports three reads (Figure 2). (1) *Regimes*. f_θ retains the six operating regimes although no loss named them: linearly decodable at 0.911 ± 0.008 , and canonically correlated with the raw settings at $\rho \approx 0.88$. This is preservation, not discovery and the regime label is k -means on three columns of the encoder’s own input, on which a linear probe already reaches 0.929 (majority class 0.301), so the read is that a purely temporal objective leaves the regime partition almost intact under a $24 \rightarrow 16$ bottleneck. (2) *Degradation*. The same embedding carries the wear axis although only the $RUL < 30$ indicator was supervised and a ridge probe to continuous RUL reaches $\rho = 0.710\pm 0.007$ where the bounds follow it, $U_{\Diamond(\text{FAIL-SOON})}$ rising $0.07 \rightarrow 0.34 \rightarrow 0.93$ by state as $L_{\Box(\text{SAFE})}$ erodes $0.93 \rightarrow 0.66 \rightarrow 0.07$. (3) *Frame structure*. The relation audits as reflexive and symmetric, and nominally transitive ($T=0.896\pm 0.007$, $B=1.000$, $4=0.998\pm 0.001$), but this is the parameterisation, not a learned frame: the accessibility is an inner-product kernel, so it is symmetric and strongly self-accessing by construction, and the implication axioms are near-vacuous. Under the diagonal-preserving null of the supplement, no axiom shows a meaningful gap.

Held-out detection is reported for completeness, not as a competitive claim: the upper bound scores AUROC 0.980 ± 0.001 against 0.990 for its own valuation head and 0.988 for logistic regression on the same sensors. The modal aggregation does not improve on the classifier it smooths; what it buys is the structure reads (1)–(2) expose. Per-state numbers and the five-axiom audit are in the supplement (*Turbofan Wear*).

5.3. Sentence Ordering (ROCStories): a learned temporal precedence relation

The temporal modality exercises the *Until* operator U , the one modal aggregator the other three never use. ROCStories Mostafazadeh et al. (2016); Sharma et al. (2018) is a standard corpus of crowd-written five-sentence everyday narratives, each a complete story with a complication and a resolution, used for commonsense story understanding; the companion Story Cloze set supplies held-out stories with a labelled correct ending. We scramble the five sentences and ask the model to recover the true order. This inverts runtime verification: there one is given the relation and asked whether a temporal formula holds on it, whereas here the formula is fixed and the relation is the unknown we learn. Each sentence is a Kripke world, embedded by a frozen sentence-transformer Reimers and Gurevych (2019); an *asymmetric* rank-32 bilinear head produces a two-place accessibility $A_\theta(w_i, w_j) = \sigma(\alpha \langle P_{\text{src}} h_i, P_{\text{tgt}} h_j \rangle + \beta)$ read as “sentence s_j plausibly follows

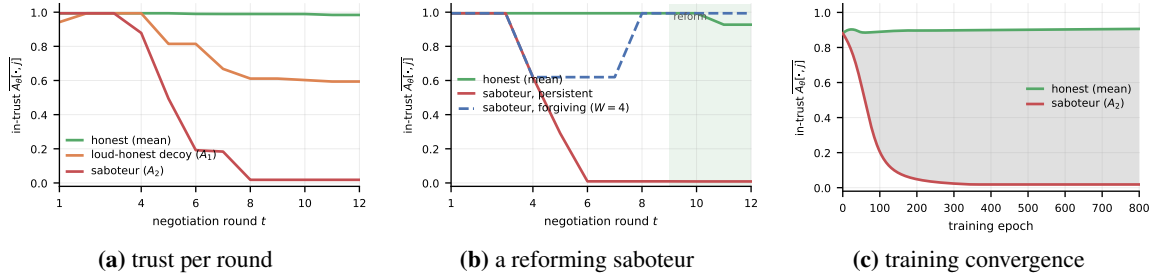


Figure 3: **The object the modal layer produces (llama3):** the in-trust $A_{\theta}[\cdot, j]$ others accord each agent. **(a)** Subtle, per round — the saboteur drains to 0.018 from round 4, honest agents hold near 0.98, the decoy, penalised only for tone, settles at 0.59; detection reads the top-3 mean, not this curve. **(b)** a saboteur backing the budget from round 9 (shaded): the default *persistent* summary stays collapsed at 0.009, a *forgiving* $W=4$ window lets trust return — one knob, same pipeline. **(c)** training: the gap reaches 90% of its final 0.89 by epoch 142.

s_i ” (24,578 trainable parameters on top of the frozen encoder). For a candidate order σ the score is a single *Until* evaluated at the first world, the adjacent-transition chain $A_{\theta}(\sigma(t-1), \sigma(t))$ must hold until the last-sentence indicator, scored over all $5! = 120$ permutations of S_5 (exact, but factorial in sequence length); training minimises the contradiction loss directly (as graph colouring does, Section 5.4), driving the true (identity) order above every scramble by a margin m .

Result and interpretation. Within a story the five sentences share characters, vocabulary and tone, so a sentence-transformer cosine ordering sits at chance (0.50/0.008); the only signal is ordinal. Over 5 seeds the bare model recovers order at pairwise 0.765 and exact 0.235 on the held-out Story-Cloze-2018 set (disjoint by text; in-domain 0.788/0.225). A stronger frozen encoder (all-mpnet-base-v2) lifts MLNN to 0.819/0.327, dead level with a prompted 7B LLM (qwen2.5-7b, 0.854/0.330) and past a same-embedding pairwise head (0.809/0.213): ordering is learnable locally but not globally; the *Until* neuron supplies that. Two structural controls confirm what carries the result: removing the contradiction margin drops pairwise to 0.708, and, most telling, tying $P_{src} = P_{tgt}$ into a direction-blind *symmetric* kernel collapses it to 0.566, barely above chance. The frame-axiom audit names the relation this task learns: a *symmetry deficit* ($B = 0.87$) and sub-reflexivity ($T = 0.78$) make it a temporal-*precedence* order, in deliberate contrast to the reflexive and symmetric relation C-MAPSS learns from the same machinery. Transitivity ($4 = 1.00$) saturates on both and separates nothing. Per-seed ablations, the encoder/LLM comparison and the full audit are in the supplementary material (*Sentence Ordering*); the supplement reads the axiom validity off four examples directly.

5.4. Graph Colouring: solving from axioms, recovering the constraint graph

Graph k -colouring is the constraint-satisfaction problem most natural to modal logic, because the adjacency relation is the accessibility relation: “adjacent nodes take different colours” is exactly the modal axiom $\bigwedge_{c=1}^k (p_c \rightarrow \neg \Diamond p_c)$, “if I am colour c , no accessible node is colour c .” We use graph colouring in both of MLNN’s two modes: *Solve*, where R is fixed and $L_{task}=0$ so a colouring is found by minimising the axiom-violation residual, and *Learn*, where the graph is hidden and a learnable A_{θ} is recovered from valid colourings.

Solve (pure satisfiability mode). The graph is given, so R is fixed and there is no task loss ($L_{task}=0$); MLNN simply drives the per-node axiom-violation residual down until no edge is monochromatic. We test on planted-3-colourable graphs in three tiers (10 graphs each; 12, 20, 30 nodes at

rising density), so each rate is out of 10 per tier and 30 overall. At that size the intervals are wide and we report them.

MLNN solves 29/30 (97%, Wilson 95% CI [0.83, 0.99]). Only one of the three comparisons is statistically supported. Against the non-modal ablation, which keeps the identical gradient solve but swaps $\square/\diamond$ for a bare quadratic peer penalty, 18/30 (60%, [0.42, 0.75]), the gap is real (Fisher $p = 0.001$): the modal operator, not the gradient solve, carries it. Against a Semantic-Loss encoding Xu et al. (2018), 26/30 (87%), the difference is *not* significant ($p = 0.35$), so we withdraw any ordering between them; and a recurrent relational-net (RRN-style) GNN ties MLNN exactly (29/30, $p = 1.0$), which is the honest reading that injecting relational structure, rather than the modal operator specifically, lifts the solve rate. Exact solvers and metaheuristics reach 100%, and supervised differentiable solvers (SATNet Wang et al. (2019), RRN Palm et al. (2018)) are strong on trained distributions. What none exposes is the per-node L_{contra} residual, which solves *from the axioms* with no training corpus and drains to zero as the colouring becomes proper (supplement, *Graph Colouring*). The full matrix is in the supplement.

Learn (inductive accessibility). The on-thesis variant hides the graph and shows the model only valid colourings. We train a learnable A_θ to satisfy the colouring axiom across all of them while pushing A_θ as large as the axiom permits; pairs that never share a colour saturate towards 1 and pairs that sometimes do are forced towards 0. From 60 proper colourings of a hidden 16-node graph the learned A_θ recovers the true adjacency at edge-recovery $AUC = 1.000$: the learned scores rank every true edge above every non-edge, so thresholding recovers the graph exactly (all 11 edges, no spurious one) given a sufficiently diverse colouring set. This is the reading a fixed-relation solver cannot give, there the rules are known in advance, and it is the paper’s central claim made literal: the accessibility relation the analyst reads is itself *learned*, and it is exactly the constraint graph.

This variant is also the paper’s one *measurable* test of interpretability: elsewhere a learned relation can only be judged by whether it looks plausible, but here the hidden graph is ground truth, so “is the artefact the analyst reads correct?” has a number, and it is exactly correct. Two further measurements, faithfulness and seed-stability ($r = 0.84$ across five seeds), are reported in the appendix.

6. Discussion and Conclusion

MLNNs extend differentiable neurosymbolic reasoning to modal logic by making (W, R, V) a first-class differentiable component with learnable valuation and accessibility. The matrix used to predict is the matrix an expert reads to audit. Across the four experiments, we showed how it can be interpreted: a trust matrix, a regime embedding with deontic bounds, a temporal-precedence order, and a recovered constraint graph. On the scalar axis MLNN is competitive on task metrics without claiming state of the art, and two results carry caveats: the doxastic margin over the LLM-judge baseline is attributable mainly to the NLI front-end, not the modal reasoning; on colouring, a relational GNN performs comparably. What survives is the artefact, since the relation computing the answer is the relation the analyst reads, and where ground truth for it exists it is recovered exactly. *Limitations.* The recurring cost is modelling: the analyst chooses the worlds and the axiom, but a poor choice may be too brittle training does not repair. Audit-regime satisfaction is also emergent and can fail silently unless measured.

Use of AI Tools: The idea is entirely human; First author used AI assistants throughout its execution for proposing and implementing experiments, literature search, drafting, and auditing our claims against the committed outputs.

References

- Samy Badreddine, Artur d’Avila Garcez, Luciano Serafini, and Michael Spranger. Logic tensor networks. *Artificial Intelligence*, 303:103649, 2022.
- Félix Bou, Francesc Esteva, Lluís Godo, and Ricardo Oscar Rodríguez. On the minimum many-valued modal logic over a finite residuated lattice. *Journal of Logic and Computation*, 21(5): 739–790, 2011.
- Xavier Caicedo and Ricardo O. Rodríguez. Standard Gödel modal logics. *Studia Logica*, 94(2): 189–214, 2010.
- Luc De Raedt, Angelika Kimmig, and Hannu Toivonen. Problog: A probabilistic prolog and its application in link discovery. In *IJCAI 2007, Proceedings of the 20th international joint conference on artificial intelligence*, pages 2462–2467. IJCAI-INT JOINT CONF ARTIF INTELL, 2007.
- Thorsten Engesser, Thibaut Le Marre, Emiliano Lorini, François Schwarzenruber, and Bruno Zanuttini. A simple integration of epistemic logic and reinforcement learning. In *Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2025)*, pages 686–694, 2025.
- Ronald Fagin, Joseph Y Halpern, Yoram Moses, and Moshe Y Vardi. *Reasoning about knowledge*. MIT press, 1995.
- Melvin C Fitting. Many-valued modal logics. *Fundamenta informaticae*, 15(3-4):235–254, 1991.
- Artur S. d’Avila Garcez, Luís C. Lamb, and Dov M. Gabbay. Connectionist modal logic: Representing modalities in neural networks. *Theoretical Computer Science*, 371(1-2):34–53, 2007. doi: 10.1016/j.tcs.2006.10.023.
- Petr Hájek. *Metamathematics of fuzzy logic*, volume 4. Springer Science & Business Media, 2001.
- Jaakko Hintikka. *Knowledge and Belief: An Introduction to the Logic of the Two Notions*. Cornell University Press, 1962.
- Karen Leung, Nikos Aréchiga, and Marco Pavone. Backpropagation through signal temporal logic specifications: Infusing logical structure into gradient-based methods. *The International Journal of Robotics Research*, 42(6):356–370, 2023.
- Ziyang Li, Jiani Huang, and Mayur Naik. Scallop: A language for neurosymbolic programming. In *Proceedings of the ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, pages 1463–1487, 2023.
- Churn-Jung Liau. Belief, information acquisition, and trust in multi-agent systems—a modal logic formulation. *Artificial Intelligence*, 149(1):31–60, 2003.
- Weilin Luo, Hai Wan, Jianfeng Du, Xiaoda Li, Yuze Fu, Rongzhen Ye, and Delong Zhang. Teaching LTL_f satisfiability checking to neural networks. In *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence (IJCAI-22)*, pages 3292–3298, 2022. doi: 10.24963/ijcai.2022/457.

- Robin Manhaeve, Sebastijan Dumancic, Angelika Kimmig, Thomas Demeester, and Luc De Raedt. DeepProbLog: Neural probabilistic logic programming. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- Nasrin Mostafazadeh, Nathanael Chambers, Xiaodong He, Devi Parikh, Dhruv Batra, Lucy Vanderwende, Pushmeet Kohli, and James Allen. A corpus and cloze evaluation for deeper understanding of commonsense stories. In *Proceedings of NAACL-HLT*, pages 839–849, 2016.
- Rasmus Berg Palm, Ulrich Paquet, and Ole Winther. Recurrent relational networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018.
- Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of EMNLP-IJCNLP*, pages 3982–3992, 2019.
- Matthew Richardson and Pedro Domingos. Markov logic networks. *Machine Learning*, 62(1-2): 107–136, 2006.
- Ryan Riegel, Naweed Khan, Haifeng Qian, Udit Sharma, Sumit Neelam, Alexander Gray, Ndivhuwo Makondo, Ronald Fagin, Shajith Ikbali, Ankita Likhyan, et al. Logical neural networks. *arXiv preprint arXiv:2006.13155*, 2020.
- Erik Rohkohl. mlsolver: Modelling of multi-agent-system as kripke structure. <https://github.com/erohkohl/mlsolver>, 2024.
- Abhinav Saxena, Kai Goebel, Don Simon, and Neil Eklund. Damage propagation modeling for aircraft engine run-to-failure simulation. In *2008 International Conference on Prognostics and Health Management*, pages 1–9. IEEE, 2008.
- Rishi Sharma, James Allen, Omid Bakhshandeh, and Nasrin Mostafazadeh. Tackling the story ending biases in the story cloze test. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 752–757, 2018.
- Yoav Shoham and Kevin Leyton-Brown. *Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations*. Cambridge University Press, 2009.
- Emile van Krieken, Erman Acar, and Frank van Harmelen. Analyzing differentiable fuzzy logic operators. *Artificial Intelligence*, 302:103602, 2022.
- Po-Wei Wang, Priya L Donti, Bryan Wilder, and J Zico Kolter. SATNet: Bridging deep learning and logical reasoning using a differentiable satisfiability solver. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, pages 6545–6554, 2019.
- Jingyi Xu, Zilu Zhang, Tal Friedman, Yitao Liang, and Guy Van den Broeck. A semantic loss function for deep learning with symbolic knowledge. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, pages 5502–5511, 2018.
- Zhun Yang, Adam Ishay, and Joohyung Lee. NeurASP: Embracing neural networks into answer set programming. In *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence (IJCAI)*, pages 1755–1762, 2020.

Modal Logic Neural Networks: Supplementary Material

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Abstract

This supplementary material accompanies the paper *Modal Logical Neural Networks*. It contains the complete theoretical analysis and proofs (Appendix A), algorithmic details of the upward–downward inference procedure (Appendix B), the scalability analysis (Appendix C), and full experimental details and ablations for all four experiments. References into the main paper are given by name (e.g. the contradiction loss, the Graph Colouring experiment), not by number.

Appendix A. Complete Theoretical Analysis

This section provides complete proofs for all theorems stated in the main paper, along with additional theoretical results on the properties of modal operators and learning dynamics.

A.1. Soundness of Differentiable Modal Operators

Theorem 1 (Soundness of Differentiable Operators). Let $x \in [0, 1]^n$. The four aggregators used by the modal neurons are one-sided in the following directions, with the stated gaps:

$$\min(x) - \tau \log n \leq \text{smin}_\tau(x) \leq \min(x), \quad \max(x) \leq \text{smax}_\tau(x) \leq \max(x) + \tau \log n, \quad (\text{S1})$$

$$\min(x) \leq \text{cp}_\tau^{\min}(x) \leq \max(x), \quad \min(x) \leq \text{cp}_\tau^{\max}(x) \leq \max(x). \quad (\text{S2})$$

Consequently $[L_\square, U_\square]$ encloses $\min(z)$ and $[L_\diamond, U_\diamond]$ encloses $\max(u)$: the operators are sound as interval-bound transformers. The pooling pair $\text{cp}_\tau^{\min}, \text{cp}_\tau^{\max}$ is what the main paper writes conv-pool_τ .

Proof. The log-sum-exp half rests on one fact about a sum of positive numbers: it is at least its largest term, and at most n copies of it. Write $x_{\min} = \min_i x_i$ and $S = \sum_i e^{-x_i/\tau}$. The largest term of S is $e^{-x_{\min}/\tau}$ (the exponential is decreasing), so

$$e^{-x_{\min}/\tau} \leq S \leq n e^{-x_{\min}/\tau}.$$

Apply $-\tau \log(\cdot)$, which is decreasing and so reverses both inequalities: $x_{\min} - \tau \log n \leq \text{smin}_\tau(x) \leq x_{\min}$. That is the first line. The smax_τ line is its mirror image, since $\text{smax}_\tau(x) = -\text{smin}_\tau(-x)$ and $\max(x) = -\min(-x)$.

The pooling operators need no computation. Each is a weighted average $\sum_i w_i x_i$ with $w_i \geq 0$ and $\sum_i w_i = 1$, and an average of numbers never leaves their range, so both sit in $[\min(x), \max(x)]$ *whatever* the weights are. This is worth saying plainly: the softmax weighting is not what makes the enclosure hold (any weights would do), so soundness here is free. What the weighting buys is *tightness*. As τ falls, the softmax concentrates on the extreme entries, and $\text{cp}_\tau^{\min}$ slides down onto $\min(x)$; the closer the smallest entry sits to its neighbours, the slower that happens, so unlike the $\tau \log n$ gap of the log-sum-exp pair this one depends on the data. $\square$

Remark (what soundness does and does not cover). Theorem 1 is about the aggregators as interval-bound transformers: fed valid intervals ($L \leq U$) that bracket the classical values, they return intervals that bracket the hard min / max, up to the gaps above. It does *not* say that every node of the formula graph keeps $L \leq U$. A modal node may emit $L_{\phi,w} > U_{\phi,w}$ when the current accessibility relation cannot be reconciled with the asserted axioms. That inversion is not a broken bound (each aggregator’s own inequality still holds) but the contradiction signal that feeds L_{contra} .

Corollary (Temperature Limits). As $\tau \rightarrow 0$, $\text{smin}_\tau(x) \rightarrow \min(x)$ and $\text{smax}_\tau(x) \rightarrow \max(x)$ uniformly at rate $\tau \log n$, and the pooling operators converge to min and max at the data-dependent rate above, so classical modal semantics is recovered in the limit. Quantitatively, holding the log-sum-exp bracket to width ε needs $\tau \leq \varepsilon / \log n$; so the temperature must be chosen against the fan-in.

Consequence: τ must shrink with the fan-in. The looseness of the bracket at one modal neuron is at most $\tau \log n$ for n aggregated neighbours, so holding the logical fidelity of the interval fixed as the frame grows requires $\tau = O(1 / \log n)$, a scaling rule rather than a tuning heuristic. Two consequences for the experiments here. First, the top- k mask is what keeps the rule affordable: with $n = k = 8$ and $\tau = 0.1$ the worst-case gap is 0.21, whereas aggregating unmasked over $|W| = 20,000$ worlds at the same τ would give a gap of 0.99 and a vacuous interval. All smin_τ and smax_τ aggregations in our implementation run over the masked neighbourhood $\tilde{A}$, so $n \leq k$ throughout and the reported bounds are the masked ones. Second, it explains the temperatures the experiments use: C-MAPSS ($\tau = 0.03$) and the colouring solver (annealed to $\tau = 0.01$) have the largest effective fan-in, and their temperatures are what the rule predicts rather than free choices.

Remark (“Is $\Box$ just a min?”). Stripped to its kernel, $\Box\phi$ is a soft minimum and $\Diamond\phi$ a soft maximum over the accessibility-gated neighbourhood, so one might object that the “modal logic” is just min/max pooling. We embrace the reduction. A universal quantifier *is* a minimum and an existential a maximum over the possible worlds, in classical and Łukasiewicz semantics alike, and $\text{smin}_\tau \rightarrow \min$ as $\tau \rightarrow 0$ (Theorem 1) is exactly the statement that the neuron is a *sound relaxation of the Kripke $\Box$* rather than an unrelated pooling. The content is not the aggregator but *what it aggregates over*: a learned, weighted, inspectable A_θ whose entries gate the implication $1 - A_{w,w'} + V_{w'}$ inside the min, so one aggregator routed through different relations gives the epistemic, doxastic, temporal and deontic readings. Two things we do *not* claim: that the bracket $[L, U]$ earns its keep on every task (on the doxastic task it does not, Proposition 2), or that an averaged frame-axiom score is evidence of structure (it is not, Lemma 1 and Table S2). “Just a min” is correct; the claim is that the min is a *faithful and inspectable Kripke $\Box$* .

A.2. De Morgan Duality of the Modal Operators

The duality $\Diamond\phi \equiv \neg\Box\neg\phi$ (with $\neg x = 1 - x$) holds for the $\text{smin} / \text{smax}$ pair by one line of algebra: in the hard limit, $1 - \min_{w'}[(1 - A_{w,w'}) + (1 - V[w', \phi])] = \max_{w'}[A_{w,w'} + V[w', \phi] - 1] = \Diamond\phi(w)$, and at finite τ the same identity follows from $\text{smax}_\tau(x) = 1 - \text{smin}_\tau(1 - x)$. The clamp to $[0, 1]$ is harmless rather than merely ignorable: since $A, V \in [0, 1]$ the inner expression $2 - A - V$ lies in $[0, 2]$, so clamping $\Box\neg\phi$ and clamping $\Diamond\phi$ agree on the boundary cases and the identity survives clamping exactly.

The pooling pair, which supplies the other endpoint of each interval, is dual too. The softmax weights are unchanged by the substitution $x \mapsto 1 - x$ up to a sign, because the constant $e^{-1/\tau}$

cancels between numerator and denominator:

$$\text{softmax}(-(1-x)/\tau)_i = \frac{e^{-(1-x_i)/\tau}}{\sum_j e^{-(1-x_j)/\tau}} = \frac{e^{x_i/\tau}}{\sum_j e^{x_j/\tau}} = \text{softmax}(x/\tau)_i.$$

Hence $\text{cp}_\tau^{\min}(1-x) = \sum_i \text{softmax}(x/\tau)_i(1-x_i) = 1 - \text{cp}_\tau^{\max}(x)$, i.e. $1 - \text{cp}_\tau^{\min}(1-x) = \text{cp}_\tau^{\max}(x)$. All four endpoints are therefore De Morgan dual in pairs, so an implementation need provide only necessity: possibility follows from the dual for *both* of its endpoints, and the contradiction residual built on either is the same object.

A.3. Convergence Analysis

Theorem 2 (Termination and exactness of inference). Fix the parameters (the accessibility A_θ and all leaf bounds) and let $G = (V, E)$ be an acyclic formula graph over the non-modal fragment $\{\neg, \wedge, \vee, \rightarrow\}$ with polarity-tracked interval semantics. Then each sweep is exact for its own system and costs $O(|V| + |E|)$ operator evaluations: one upward sweep in topological order computes the exact fixed point of the upward rules, and one downward sweep in reverse topological order the exact fixed point of the downward rules given its parent bounds. The *joint* fixed point of the two systems is reached by iterating the pair, which terminates because every update is a pure tightening of a bounded quantity. The guarantee concerns inference, not training: each gradient step changes A_θ , and the next forward pass reruns inference at the new parameters.

Proof. Interval reasoning needs every operator to be *monotone*: widening an input must never narrow an output. Two of the Łukasiewicz operators fail that test as written: negation $1-x$ and the antecedent slot of $\min(1, 1-a+b)$ both run backwards, so a larger input gives a smaller output. The standard remedy is to swap the endpoints wherever an argument enters negatively:

$$\neg[L, U] = [1-U, 1-L], \quad [L_a, U_a] \rightarrow [L_b, U_b] = [\min(1, 1-U_a+L_b), \min(1, 1-L_a+U_b)].$$

With that convention every operator is monotone in the componentwise order ($[L_1, U_1] \sqsubseteq [L_2, U_2]$ iff $L_1 \leq L_2$ and $U_1 \leq U_2$), which is the usual correctness condition for interval abstract interpretation, and it is exactly what the downward rule for $\neg$ implements.

The rest is acyclicity, and it needs no limiting argument at all. In topological order a node is reached only once every one of its inputs is already final, so a single upward pass computes each node exactly, once; the downward pass does the same in reverse order. Each pass is a *scheduled* walk over a finite DAG, not an iteration towards a limit, which is why the cost is a count of nodes and edges rather than a convergence rate. Iterating the two passes terminates as well: every update either raises an L or lowers a U within $[0, 1]$, and a bounded quantity cannot be pushed the same way for ever. $\square$

Remark (why one round is not enough). One upward sweep plus one downward sweep does *not* in general give the joint fixed point, and the reason is the very case the downward pass exists for. Take leaves a, b, c , all at $[0, 1]$, with $\phi_1 = a \wedge b$ asserted $[1, 1]$ and $\phi_2 = a \vee c$. The upward sweep evaluates ϕ_2 while a is still $[0, 1]$ and gets $[0, 1]$; the downward sweep then fires $L_a \leftarrow \max(L_a, L_{\phi_1}) = 1$. Now ϕ_2 is stale (with $a = [1, 1]$ its lower bound is 1), but the sweep is over. In general any node with more than one parent can be tightened after a sibling has already consumed it, and sharing subformulae across asserted axioms is precisely what the downward pass is for. The released implementation therefore iterates to a tolerance and warns if its iteration budget runs out first.

Remark (Gradient flow of the contradiction loss). The contradiction loss

$$L_{\text{contra}} = \sum_{w,\phi} \max(0, L_{\phi,w} - U_{\phi,w})$$

is subdifferentiable everywhere and differentiable wherever $L_{\phi,w} \neq U_{\phi,w}$, with gradient supported on the contradicted nodes ($L_{\phi,w} > U_{\phi,w}$). This is immediate: $t \mapsto \max(0, t)$ has derivative 1 for $t > 0$, 0 for $t < 0$ and subdifferential $[0, 1]$ at $t = 0$, and a sum of subdifferentiable functions is subdifferentiable, so the only non-differentiable points are nodes sitting exactly at $L = U$. Consistent nodes contribute no gradient at all, and optimisation acts only where an axiom is actually violated.

A.4. Structural Properties of Learnable Accessibility

The expressiveness of A_θ follows trivially for the direct parameterisation $A_\theta = \sigma(M_\theta)$, since σ is a bijection $\mathbb{R} \rightarrow (0, 1)$. The more interesting question is whether gradient-based learning of A_θ can *recover and preserve* structural properties that correspond to classical modal axiom systems. The classical axioms each name a structural condition on the accessibility relation R , with a domain reading that depends on the modality (Table S1); a bare numeral or letter names a single axiom, and concatenations name combined systems (KT, S4 = K+4, S5 = T+4+B, KD).

Table S1: The five frame axioms on R : logical form, structural condition, domain reading, and the modality each one distinguishes. The same $\Box$ neuron yields knowledge (K_a : T, 4, 5 = S5), belief (B_a : 4, 5, D = KD45), obligation (O : D = KD) and time (G : 4, D over a linear order) according to which axioms its relation satisfies, so factivity is a fact about the relation, not about the operator. MLNN can *impose* any of them as a penalty (Theorem 3) or *measure* them by audit (below).

Axiom	Form	Condition on R	Reading	Held by
T (reflexive)	$\Box\phi \rightarrow \phi$	$R(w, w)$	epistemic factivity	K_a
4 (transitive)	$\Box\phi \rightarrow \Box\Box\phi$	$R(u, v) \wedge R(v, w) \Rightarrow R(u, w)$	positive introspection	K_a, B_a, G
B (symmetric)	$\phi \rightarrow \Box\Diamond\phi$	$R(u, v) \Rightarrow R(v, u)$	mutual reachability	none
5 (Euclidean)	$\Diamond\phi \rightarrow \Box\Diamond\phi$	$R(u, v) \wedge R(u, w) \Rightarrow R(v, w)$	negative introspection	K_a, B_a
D (serial)	$\Box\phi \rightarrow \Diamond\phi$	every u has a successor	no dead-ends	B_a, O, G

Theorem 3 (Structural Recoverability). Let $\mathcal{P}$ be one of reflexivity, transitivity, symmetry, the Euclidean property or seriality (axioms T, 4, B, 5, D). For each there is a differentiable regulariser $R_{\mathcal{P}}(A) \geq 0$ whose zero set is the matrices satisfying $\mathcal{P}$, and training uses $L = L_{\text{task}} + \gamma R_{\mathcal{P}}$. Two of the five carry a qualification: under the sigmoid parameterisation T is approached but never attained, and D needs the debiasing given below.

Construction. For 4, B and 5 the construction is the same one every time: *write down the property as an inequality, and penalise how far it is violated*. Transitivity in the product t-norm asks $A_{wv}A_{vw'} \leq A_{ww'}$; the Euclidean condition asks $A_{uv}A_{uw} \leq A_{vw}$; symmetry asks $A_{uv} = A_{vu}$. Squaring the shortfall gives

$$\begin{aligned}
 R_{\text{trans}} &= \sum_{w,v,w'} \max(0, A_{wv}A_{vw'} - A_{ww'})^2, \\
 R_{\text{eucl}} &= \sum_{u,v,w} \max(0, A_{uv}A_{uw} - A_{vw})^2, \\
 R_{\text{sym}} &= \|A - A^\top\|_F^2.
 \end{aligned}$$

Each is a sum of non-negative terms, so it vanishes exactly when every term does, which is exactly the property, and such matrices exist in the reachable parameter space, so the zero is attainable. That is the whole argument for these three, and it is worth being blunt that it is close to a tautology: the penalty *is* the property’s residual, so “its zero set is the property” carries little information beyond well-posedness. It says nothing about whether gradient descent finds those zeros. (Note the t-norm. These compose A with the *product* $A_{uv}A_{vw}$, whereas the frame-axiom audit of Section A.5 uses Łukasiewicz, $\max(0, A_{uv} + A_{vw} - 1)$. Product dominates Łukasiewicz pointwise on $[0, 1]^2$, so $R_{\text{trans}}(A) = 0$ implies $S_4(A) = 1$ but not conversely: the regulariser enforces strictly more than the audit measures. We use product in the regulariser because its gradients do not vanish on half the unit square, and Łukasiewicz in the audit because it matches the connectives the forward pass evaluates; substituting the Łukasiewicz composition into R_{trans} makes the two coincide, and the argument above goes through verbatim.) Symmetry is the one of the three with a genuine consequence, proved below: a pointwise-symmetric A validates the Brouwerian axiom $\phi \rightarrow \Box\Diamond\phi$ in the graded reading.

The two remaining cases are not definitional, and are where the content sits.

1. **Reflexivity (T)** is $R_{\text{refl}}(A) = \sum_w (1 - A_{ww})^2$, and the subtlety is that the trainable parameter is M_θ , not A , with $A = \sigma(M_\theta)$. Differentiating in the right coordinates, stationarity reads

$$\sigma'(M_{ww}) [\nabla_{A_{ww}} L_{\text{task}} - 2\gamma (1 - A_{ww})] = 0,$$

which has two branches. Away from saturation ($\sigma'(M_{ww}) \neq 0$) the bracket must vanish, giving $A_{ww} = 1 - \frac{1}{2\gamma} \nabla_{A_{ww}} L_{\text{task}}$, so $A_{ww} \rightarrow 1$ as $\gamma \rightarrow \infty$ provided L_{task} ignores the diagonal. But σ maps onto the *open* interval $(0, 1)$, so $A_{ww} = 1$ is a limit and never a stationary point; and the other branch, $\sigma'(M_{ww}) \rightarrow 0$, satisfies stationarity vacuously by saturating. Exact reflexivity therefore needs the hard regime (pinning the diagonal), not the soft one.

2. **Seriality (D)** needs care because the obvious surrogate is *wrong*. Writing it with the smooth maximum, $\sum_u \max(0, 1 - \text{smax}_{\tau,v} A_{uv})^2$ vanishes on relations that are not serial at all: smax_τ over-estimates $\max$ by up to $\tau \log n$ (Theorem 1), so any row with $\max_v A_{uv} \geq 1 - \tau \log n$ already scores zero: at $\tau = 0.1$, $n = k = 8$ that admits a row whose true maximum is 0.79. Subtracting the known bias restores the “iff”:

$$R_{\text{ser}}(A) = \sum_u \max(0, 1 - \text{smax}_{\tau,v} A_{uv} + \tau \log n)^2.$$

This is conservative because $\text{smax}_\tau(x) - \tau \log n \leq \max(x)$, so $R_{\text{ser}} = 0$ forces $\max_v A_{uv} = 1$: every world has a successor, no dead-ends, and $\Box\phi \rightarrow \Diamond\phi$ holds. As with T, under $A = \sigma(M)$ this is attained only in the limit. Seriality is also the one axiom whose penalty never fights the task gradient, since one high-weight successor per row leaves every other entry free.

Proof that symmetry validates B. The crisp argument (“ $A_{ww'} > 0$ implies $A_{w'w} > 0$ ”) is vacuous here, because σ makes every entry strictly positive for *every* matrix, symmetric or not. The graded statement does hold, and directly. Work in the hard limit and fix w . For any v , choosing $u = w$ inside the inner maximum gives

$$\Diamond\phi(v) = \max_u [A_{vu} + \phi(u) - 1] \geq A_{vw} + \phi(w) - 1 = A_{wv} + \phi(w) - 1,$$

the last step by symmetry. Substituting into the outer minimum, the A_{wv} terms cancel:

$$\Box\Diamond\phi(w) = \min_v [(1 - A_{wv}) + \Diamond\phi(v)] \geq \min_v [(1 - A_{wv}) + A_{wv} + \phi(w) - 1] = \phi(w),$$

which is $\phi \rightarrow \Box\Diamond\phi$ in the Łukasiewicz reading. Note this uses symmetry pointwise on the weights, not merely on the support. $\square$

Regimes. Adding a regulariser cannot violate Theorem 1, which is a pointwise inequality about the forward aggregators quantified over all $A_\theta \in [0, 1]^{|W| \times |W|}$: an additive loss term changes neither the functional form of $\Box/\Diamond$ nor the range of A_θ . What it does change is where an (axiom, parameterisation) pair sits among three regimes. *Hard*: the property is guaranteed by construction, e.g. pinning $A_{ww} = 1$ for reflexivity, or a fixed temporal-lag mask making the relation a strict pre-order. *Soft*: a regulariser $R_{\mathcal{P}}$ (Theorem 3) pulls A_θ toward the property without forcing it. *Audit*: the property is unconstrained during training and measured post-hoc. The three trade enforcement against expressivity: a hard constraint cannot be contradicted by the data, whereas an audit reveals whatever structure the data induced. Only the hard regime attains the axioms exactly; under $A = \sigma(M)$ the soft regime merely approaches them. Emergent satisfaction can fail silently, which is why we measure it.

A.5. Frame-axiom audit

Because the learned A_θ is the same matrix the forward pass uses, each frame axiom can be scored directly on it as an average Łukasiewicz satisfaction degree $S_{\mathcal{P}}(A) \in [0, 1]$ (1 = satisfied, 0 = systematically violated), using the Łukasiewicz implication $\text{impl}(a, b) = \min(1, 1 - a + b)$ and conjunction $a \wedge b = \max(0, a + b - 1)$:

$$\begin{aligned} S_T(A) &= \text{avg}_u A_{uu}, & S_4(A) &= \text{avg}_{u,v,w} \text{impl}(A_{uv} \wedge A_{vw}, A_{uw}), \\ S_B(A) &= \text{avg}_{u,v} (1 - |A_{uv} - A_{vu}|), & S_5(A) &= \text{avg}_{u,v,w} \text{impl}(A_{uv} \wedge A_{uw}, A_{vw}), \end{aligned}$$

with seriality read off the row maxima, $S_D(A) = \text{avg}_u \max_v A_{uv}$.

Vacuity. The two implication axioms, S_4 and S_5 , are easy to satisfy for the wrong reason. Łukasiewicz conjunction sends the antecedent to 0 whenever $A_{uv} + A_{vw} \leq 1$, and $\text{impl}(0, b) = 1$ for every b , so any triple with two weak links counts as satisfied whatever the third link is. Sparse or low-variance relations therefore score near 1 on S_4 and S_5 whatever their structure. This is not hypothetical: the doxastic objective (Appendix D.1) pushes every row toward a common profile, so the learned trust matrix is close to column-constant, the exact shape that triggers the degeneracy.

Lemma 1 (Vacuity on column-constant relations) *If A is column-constant, $A_{uv} = c_v$ for all u with $c \in [0, 1]^{|W|}$, then $S_4(A) = S_5(A) = 1$ for every c , and $S_B(A) = \text{avg}_{u,v} (1 - |c_v - c_u|)$.*

Proof For S_4 the summand is $\text{impl}(\max(0, c_v + c_w - 1), c_w)$: if $c_v + c_w \geq 1$ it equals $\min(1, 2 - c_v) = 1$, and if $c_v + c_w < 1$ the antecedent is 0 so it is again 1. S_5 gives the same expression, and the S_B formula follows from $A_{uv} - A_{vu} = c_v - c_u$. ■

Correction. A single average cannot separate real structure from vacuity, so for the two implication axioms (4 and 5) we report three numbers instead of one. (i) The coverage $\kappa_{\mathcal{P}}$ is the fraction of triples that actually test the axiom, i.e. those with a non-zero antecedent; low coverage means almost every triple was vacuous, so the average on its own says little. (ii) The non-vacuous score $S_{\mathcal{P}}^{\text{nv}}$ is the satisfaction averaged over those testing triples only, and is left undefined when $\kappa_{\mathcal{P}} = 0$ rather than reported as a meaningless 1. (iii) The null baseline $S_{\mathcal{P}}^{\text{null}}$ is the score expected from the relation’s shape alone: the null hypothesis is that the relation has no genuine transitive (or Euclidean) structure, so that its score is fixed by how strong each world’s outgoing links are, not by which worlds they point to. We estimate it by re-scoring 100 random copies of A in which the weights in each row are shuffled onto different targets, keeping each world’s self-weight, its set of out-weights and its strongest link unchanged. An axiom is credited only when S^{nv} clearly exceeds S^{null} at non-negligible coverage; otherwise the apparent satisfaction is only an artefact of shape. Reflexivity (T) and seriality (D) need none of this: they are plain averages, not implications, and the shuffling leaves them unchanged.

Table S2: Frame-axiom audit of the three learned relations. The T and D columns (grouped “plain”) are the mean reflexivity and seriality, which need no correction. For symmetry (B), transitivity (4) and the Euclidean axiom (5), S is the mean satisfaction, shown next to the coverage κ (fraction of triples that test the axiom), the non-vacuous score S^{nv} (mean over those triples only), and the null baseline S^{null} (the same score on 100 shape-matched random relations; per-relation s.d. ≤ 0.02). An axiom counts as genuinely present only when S^{nv} exceeds S^{null} . Doxastic is the mean over the 15 Subtle runs; C-MAPSS is 120 engine blocks of the temporal relation; colouring is the 16-node adjacency recovered from proper colourings alone; ROCStories is 1,571 Cloze stories over 5 seeds. Reproduced by the released audit scripts.

relation	plain		B (symmetric)		4 (transitive)				5 (Euclidean)			
	T	D	S	S^{null}	κ	S	S^{nv}	S^{null}	κ	S	S^{nv}	S^{null}
doxastic trust	1.00	1.00	0.687	0.736	0.94	1.000	1.000	0.937	0.93	0.978	0.976	0.930
colouring (learned)	0.00	0.51	1.000	0.885	0.01	0.996	0.637	0.669	0.01	0.996	0.637	0.655
C-MAPSS temporal	0.14	0.14	1.000	0.961	0.00	1.000	<i>und.</i>	<i>und.</i>	0.00	1.000	<i>und.</i>	<i>und.</i>

Results. Read this way, the axioms that clear their null are the symmetry ones, in both directions. Graph colouring is genuinely symmetric ($S_B = 1.000$ against a null of 0.885), as an undirected graph must be, and correctly irreflexive ($S_T = 0$). Two other relations sit *below* their symmetry null, the signature of a preferred direction: the doxastic trust matrix (0.687 against 0.736), whose directed in-trust/out-trust asymmetry the experiment is about, and the ROCStories order (0.836 against 0.864), whose asymmetry is oriented forward in narrative time.

The implication axioms (transitivity 4, Euclidean 5) clear their null nowhere by a meaningful margin, for different reasons. For the *C-MAPSS* temporal relation every weight is at most its diagonal value of 0.14, so $A_{uv} + A_{vu} \leq 1$ for every triple, coverage is zero, and S^{nv} is undefined:

an $S_4 = 1.00$ here carries no information at all. For *graph colouring* coverage is almost zero ($\kappa_4 = 0.01$), and on the few covered triples the score drops to $S_4^{\text{nv}} = 0.637$, below its null of 0.669; the same audit on the ground-truth graph gives $S_4^{\text{nv}} = 0.000$ against an averaged $S_4 = 0.988$, confirming that an undirected graph is generically non-transitive even where the plain average suggests otherwise. For *doxastic trust* coverage is high ($\kappa_4 = 0.94$) and $S_4^{\text{nv}} = 1.000$ is real, but a relation with the same out-degrees and weights already scores 0.937, so all but 0.06 is shape, exactly what Lemma 1 predicts. *ROCStories* is the lone exception: there the axioms are genuinely non-vacuous (coverage $\kappa_4 = 1.00$, against 0.00 for C-MAPSS) and satisfied ($S_4^{\text{nv}} = 1.000$), yet a shape-matched order is almost as transitive ($S_4^{\text{null}} = 0.996$), so the gap is only +0.004: an order is transitive, but so is its shape.

Interpretation. The verdicts above require three qualifications, after which the four experiments fall into a clear pattern.

- **Doxastic trust.** Reflexivity here is imposed, not learned: every agent’s self-access is fixed at 1 ($S_T = 1.00$ at zero variance across runs, and the training prior acts only off the diagonal). Reflexivity is exactly the axiom (T) that separates a knowledge frame (S5) from a belief frame (KD45), so a fixed unit diagonal reflects the modelling assumption that an agent accesses its own world, not a discovered property. The learned content is the off-diagonal trust, and the symmetry deficit ($S_B = 0.687$ below its null of 0.736) shows that this trust is directed: the saboteur is identified by the loss of trust directed *towards* it, not by its own fixed self-trust.
- **C-MAPSS.** The accessibility is an inner-product kernel, so it is symmetric and strongly self-accessing by construction rather than by learning, and the implication axioms are vacuous on the sparse relation. The higher diagonal of the contradiction-trained relation (0.90 against 0.14 for a fit to temporal adjacency) follows its greater density and shows no gap under a diagonal-preserving null. No classical frame is recovered; the substantive C-MAPSS result is the monotone behaviour of the truth bounds over an engine’s life, where the necessity of “safe” falls and the possibility of “fail-soon” rises, which the frame audit does not measure.
- **ROCStories.** Its content is an order, and the audit reflects that. Unlike C-MAPSS its implication axioms are non-vacuous (coverage ≈ 1) and satisfied, but a shape-matched order is almost as transitive, so the decisive result is the directedness: symmetry below its null (0.836 against 0.864), oriented forward in narrative time. This is the directional analogue of the doxastic deficit, and it is what the task needs: constraining the relation to be symmetric lowers exact-order accuracy from 0.23 to 0.08 (Appendix D.3).

The pattern is consistent. Where the question is which symmetries a relation carries, the audit is decisive: colouring comes out symmetric, while the doxastic trust matrix and the ROCStories order come out directed, each sitting below its symmetry null. The implication axioms are where care is needed: vacuous (C-MAPSS, colouring), shape-explained (doxastic), or non-vacuous but no stronger than shape (ROCStories), so no relation earns a transitivity claim from this audit. C-MAPSS is the one relation the frame audit leaves empty-handed: its structure lives in the truth bounds over an engine’s life, not in the relation.

Takeaway. The audit turns the learned relation into a named modal-logic object: a domain expert can ask whether the recovered matrix is approximately an equivalence relation (S5), a directed

preorder (S4), or neither, and can localise where the structure breaks by reading the per-tuple satisfaction. It is post-hoc and adds no training signal. In our experiments we default to this audit regime: we impose no structural regularisers, let A_θ learn from L_{contra} and any task loss, and read the recovered structure off the matrix. When domain knowledge instead mandates a specific axiom system, Theorem 3 guarantees it can be enforced, hard or soft, without sacrificing differentiability or soundness.

Appendix B. Details of Algorithms

B.1. Upward-Downward Inference Algorithm

The core inference procedure in MLNNs propagates truth bounds through the logical formula graph in one upward and one downward sweep (Theorem 2). The algorithm is described as follows:

Given a formula graph $G = (V, E)$, an accessibility matrix A_θ and initial bounds (all unknown formulae at $[0, 1]$), inference returns tightened bounds. The **upward pass** visits $\phi \in V$ in topological order, leaving atoms alone and evaluating each compound from its children: the propositional connectives by the Łukasiewicz rules (e.g. $L_{\psi_1 \wedge \psi_2} = \max(0, L_{\psi_1} + L_{\psi_2} - 1)$, $U = \min(U_{\psi_1}, U_{\psi_2})$), and the modal nodes by the four aggregators of Theorem 1 in the directions fixed there,

$$\begin{aligned} L_{\Box\psi,w} &= \text{smin}_\tau((1 - \tilde{A}_{w,w'}) + L_{\psi,w'}), & U_{\Box\psi,w} &= \text{cp}_\tau^{\min}((1 - \tilde{A}_{w,w'}) + U_{\psi,w'}), \\ L_{\Diamond\psi,w} &= \text{cp}_\tau^{\max}(\tilde{A}_{w,w'} + L_{\psi,w'} - 1), & U_{\Diamond\psi,w} &= \text{smax}_\tau(\tilde{A}_{w,w'} + U_{\psi,w'} - 1). \end{aligned}$$

The **downward pass** then visits V in reverse topological order, applying the inverse updates below to tighten child bounds. Every update only raises an L or lowers a U . All aggregations are taken over the masked neighbourhood $\tilde{A}$ (Section B.3), so the fan-in is $n \leq k$ and the $\tau \log n$ gap of Theorem 1 is bounded by $\tau \log k$.

Atomic and non-modal compound nodes maintain the invariant $L_{\phi,w} \leq U_{\phi,w}$ by construction, under the polarity convention of Theorem 2; modal compound nodes are exempt, for the reason given in the soundness-scope remark of Section A.

Inverse updates. The downward pass visits formulae in reverse topological order and applies the following inverse updates, each a pure tightening (lower bounds only increase, upper bounds only decrease) of the child bounds given the parent bounds $[L_\phi, U_\phi]$:

$$\begin{aligned} \phi = \neg a : & \quad L_a \leftarrow \max(L_a, 1 - U_\phi), & U_a & \leftarrow \min(U_a, 1 - L_\phi), \\ \phi = a \wedge b : & \quad L_a \leftarrow \max(L_a, L_\phi), & U_a & \leftarrow \min(U_a, \min(1, U_\phi + 1 - L_b)), \\ \phi = a \vee b : & \quad L_a \leftarrow \max(L_a, \max(0, L_\phi - U_b)), & U_a & \leftarrow \min(U_a, U_\phi), \\ \phi = a \rightarrow b : & \quad L_b \leftarrow \max(L_b, \max(0, L_\phi + L_a - 1)). \end{aligned}$$

The negation update is exact, since Łukasiewicz negation is an involution. The implication update is Łukasiewicz modus ponens and is sound: $\phi \geq L_\phi$ and $a \geq L_a$ give $b \geq \phi + a - 1 \geq L_\phi + L_a - 1$; no antecedent update (modus tollens) is applied. The conjunction updates, applied symmetrically to b , are sound for the Łukasiewicz reading $\phi = \max(0, a + b - 1)$: $\phi \leq a$ gives $a \geq L_\phi$, and $a \leq \phi + 1 - b \leq U_\phi + 1 - L_b$; asserting a conjunction true ($[L_\phi, U_\phi] = [1, 1]$), the dominant use case, therefore propagates $L = 1$ to both conjuncts. The disjunction updates, applied symmetrically to b , are sound for $\phi = \min(1, a + b)$: when $U_\phi < 1$ the clamp is inactive and $a = \phi - b \geq L_\phi - U_b$,

while $\phi \geq a$ always gives $a \leq U_\phi$; when $U_\phi = 1$ the lower update degenerates to no-op, as it should. Only the modal nodes $\Box$, $\Diamond$ and *Until* receive no downward update, and the reason is specific to them: inverting an aggregation over $\tilde{A}$ admits no canonical per-world factorisation, since many valuation profiles over the neighbourhood produce the same aggregate. An incompatibility between a modal node and an assertion on it surfaces instead as the bound inversion that feeds L_{contra} . The downward updates become active when several axioms share subformulae and bounds must be completed across them, which no experiment here does.

B.2. Gradient Computation and Backpropagation

For the $\Box$ lower bound the gradient with respect to the accessibility matrix is

$$\frac{\partial L_{\Box\phi,w}}{\partial A_{\theta,w,w'}} = - \frac{\exp(-(1 - A_{\theta,w,w'} + L_{\phi,w'}) / \tau)}{\sum_{w''} \exp(-(1 - A_{\theta,w,w''} + L_{\phi,w''}) / \tau)}$$

which is the softmax weight of world w' in the smooth minimum. Note that the weight is on a small *argument* $z_{w'} = (1 - A_{\theta,w,w'}) + L_{\phi,w'}$, not on a small truth value: a world that is barely accessible ($A_{\theta,w,w'} \approx 0$) has large $z_{w'}$ and receives essentially no gradient even if $L_{\phi,w'} = 0$. The update therefore concentrates on worlds that are *both* accessible and violating, which is precisely the pair jointly responsible for a contradiction.

B.3. Masking and Sparsity

For computational efficiency with large world sets, we implement top- k masking where each world only considers the k most accessible neighbours. The masked accessibility matrix is

$$\tilde{A}_{w,w'} = \begin{cases} A_{\theta,w,w'} & \text{if } w' \in \text{top-}k(A_{\theta,w,:}) \\ 0 & \text{otherwise} \end{cases}$$

This reduces the cost of *each modal aggregation* from $O(|W|^2)$ to $O(k \cdot |W|)$ once the mask is given. Obtaining the mask is a separate matter: computing $\text{top-}k(A_{\theta,w,:})$ exactly requires scoring all $|W|$ candidates per row, so the end-to-end cost of our current implementation remains $O(|W|^2)$, consistent with the quadratic memory and time we measure in Table S3. Under the metric parameterisation the mask can in principle be retrieved in sub-quadratic time by approximate maximum-inner-product search over the embeddings; we do not do this, and we flag it as the change required before the $O(k \cdot |W|)$ figure describes the whole pipeline rather than one operator. The ring ablation (Table S3, right) shows structure recovery is stable for $k \in [4, 16]$, with $k = 8$ the default.

Core API. All four experiments are a few lines over the $[0, 1]$ matrix A : only the relation and the formula change, the operators are differentiable bound-propagators, and the loss is always a contradiction residual (the temporal case adds `F.until`).

```
from torchmodal import functional as F
box = F.necessity(prop, A, tau=0.1)      # [ ]phi : smin over neighbours
dia = F.possibility(prop, A, tau=0.1)    # <>phi : smax over neighbours
# colouring p_c -> ~<>p_c (per-node contradiction residual):
viol = (1 - F.implies(pc, F.negation(F.possibility(pc, A)))) .clamp(min=0)
# doxastic trust: maximise [ ]_i(coherent), zero set shared with Prop. F.1's
# residual
```

```

loss = ((1 - F.necessity(consistent, A, tau=0.1)) ** 2).mean()
# deontic (C-MAPSS): safe MUST hold early, fail-soon MAY hold late
U_fail = F.possibility(fail, A)[: , 1]; L_safe = F.necessity(safe, A)[: , 0]
loss = ((1 - U_fail)**2)[rul < 30].mean() + (U_fail**2 + (1 - L_safe)**2)[rul >
    100].mean()
# temporal (ROCStories): one Until at the first world scores a sentence order
score = F.until(follows, is_last, chain, tau=0.1)[0]

```

The same operators back the higher-level building blocks, where only the accessibility parameterisation is swapped, never the logic:

```

from torchmodal.nn import LearnableAccessibility, MetricAccessibility
from torchmodal.systems import MultiAgentKripke
A = LearnableAccessibility(num_worlds, top_k=8) # dense [0,1]
logits
A = MetricAccessibility(num_worlds=num_worlds, embed_dim=64) # scalable, sub-
    quadratic
kripke = MultiAgentKripke(num_agents=6, num_steps=5) # the doxastic
    trust model
resid = F.contradiction(bounds) # sum of relu(L - U): the only training
    signal

```

Appendix C. Scalability Analysis: Synthetic “Diplomacy” Ring

The synthetic “Diplomacy ring” serves as both a diagnostic benchmark and a scalability testbed. We construct a minimal multi-agent environment where the ground-truth accessibility structure is known, allowing us to directly test whether MLNNs can recover it purely from logical constraints, and measure how performance scales with the number of agents.

Methodology We construct a synthetic environment governed by a known ground-truth Kripke structure in the form of a directed ring graph. The scenario consists of N agents arranged in a ring, where each Agent i observes a specific set of facts but requires a “Beacon” that is exclusively possessed by Agent $i + 1$. The model is trained on two specific modal constraints: Consistency ($\Box$), which dictates that an agent must not trust neighbours who contradict their direct observations; and Expansion ($\Diamond$), which requires an agent to trust at least one neighbour who possesses the necessary Beacon. Accuracy (“Acc” in Table S3) is the fraction of rows of the learned A_θ whose arg max is the true ring successor $i + 1$; MSE is against the ground-truth ring adjacency. The *Fixed R* control replaces the learned relation by the all-ones (fully connected) relation with V alone trainable, so its MSE of 0.25 is that of a uniform 0.5 predictor against a $\{0, 1\}$ target: it is a floor, not a competitive baseline.

Results. To empirically validate the computational complexity bounds (the complexity proposition in the main paper), we evaluated the MLNN on the synthetic ring task at $|W| \in \{200, 10,000, 20,000\}$. We compared two parameterisations. The standard Dense form is a full $|W| \times |W|$ matrix of learnable logits, exact but quadratic in memory ($4|W|^2$ bytes at float32) and in gradient state. The Metric form is $\hat{A}_\theta(i, j) = \sigma(\mathbf{h}_i^{\text{src}\top} \mathbf{h}_j^{\text{tgt}} + b)$, with *separate* source and target embeddings of dimension $d = 64$, i.e. $2d|W|$ parameters. The two embeddings are not an implementation detail: a single shared embedding would force $A_\theta = A_\theta^\top$ and could represent neither the directed ring nor a temporal precedence.

The results (Table S3) confirm the parameter bottleneck of the Dense approach. At $N = 10,000$ the Dense model instantiates 100 million parameters against the Metric model’s 1.28 million, and

Table S3: Analysis of the Synthetic Diplomacy Ring. **Left:** Scalability of Metric vs. Dense parameterisation up to 20k worlds; note that memory and time grow $\approx$ quadratically for *both* modes, so the metric head buys a constant factor rather than a better exponent. **Right:** Robustness of the model ($N = 20$) to hyperparameters and the necessity of learning A_θ . The task is saturated (all accuracies 100%, MSE flat to one significant figure), so this table demonstrates insensitivity on an easy problem rather than robustness in general.

N	Mode	Params	Mem (MB)	Time	Acc	Ablation	Value	MSE
200	Dense	40k	17.8	0.9s	100%	Temp (τ)	0.05	4e-4
	Metric	25k	17.6	1.2s	100%		0.1 (Def)	4e-4
10k	Dense	100M	3,836	114s	100%		0.2	4e-4
	Metric	1.2M	2,705	106s	100%	Mask (k)	4	7e-4
20k	Dense	<i>skipped (projected out-of-memory)</i>					8 (Def)	4e-4
	Metric	2.5M	10,728	428s	100%		16	3e-4
						Relation	Fixed R	0.25
							Learned	4e-4

at $N = 20,000$ we skip the Dense run because its 400 million parameters plus several $N \times N$ activation tensors project past the 16 GB device, while the Metric model remains trainable on the same single T4 GPU (consuming 10.7 GB).

We are careful not to over-read this. The saving is in *parameters and gradient state*, which is what removes the Dense cliff; it is not an asymptotic improvement in the resources the table measures. Between $N = 10,000$ and $N = 20,000$ the Metric mode’s memory grows by $3.97\times$ and its time by $4.04\times$ for a $2\times$ increase in N : both are $O(N^2)$ to two significant figures, as expected given that the exact top- k mask is still formed by scoring all pairs (Section B.3). Extrapolating the measured curve, the Metric mode would itself exhaust the same 16 GB device at roughly $N \approx 25,000$. The defensible claim is that the metric head buys about an order of magnitude in parameters and a constant factor in memory, deferring rather than removing the quadratic wall; removing it requires the approximate-retrieval mask discussed in Section B.3.

Embedding. The metric form buys something the dense matrix cannot: forcing accessibility through geometry makes the embedding itself an inspectable artefact alongside A_θ . We had read C-MAPSS as the concrete case: the same f_θ , trained only against a temporal-locality target with no regime supervision, organising its space by operating regime, linearly decodable at 0.921. The control is decisive against that reading. There are six regimes, sized 14,551/7,247/3,423/7,180/3,778/12,120 over the 48,299 training worlds, so the floor is the majority-class rate 0.301, not chance 0.167; and the regime label is k -means on the three raw op-setting columns, which are *part of the encoder’s input*. A five-fold linear probe reaches 0.929 on the encoder input, 0.929 on the three op-settings alone, and 0.929 on the 21 sensor channels with the op-settings deleted, the sensors being strongly regime-dependent. Against that ceiling the embedding’s 0.921 is a small *loss* of information already present, not a discovery of structure that was not. The defensible statement is the weaker one: f_θ *preserves* the regime partition almost intact under a $24 \rightarrow 16$ bottleneck trained on a purely temporal objective. We keep the t -SNE panel as an illustration of that preservation and withdraw the emergence reading.

Defaults. Unless an experiment states otherwise (C-MAPSS uses $\tau = 0.03$; the colouring solver anneals τ from 0.5 to 0.01), we default to temperature $\tau = 0.1$, contradiction weight $\beta = 0.1$, and top- k mask $k = 8$; the ring ablation (Table S3, right) shows structure recovery is stable across $\tau \in [0.05, 0.2]$ and $k \in [4, 16]$ on that (saturated) task. The lower temperatures elsewhere follow the fan-in rule $\tau = O(1/\log n)$ of Appendix A rather than per-task tuning.

Appendix D. Experiment Details

This appendix collects the per-example detail deferred from the Experiments section of the main paper, one subsection per experiment: the doxastic saboteur, C-MAPSS turbofan wear, ROCStories sentence ordering, and graph colouring. All four examples are reproducible end-to-end, each emitting a machine-readable summary of the numbers quoted in the body.

D.1. Doxastic Deception (LLM-swarm saboteur)

Setup. Six agents, each driven by the same locally-served LLM backbone (via *ollama*), negotiate a \$50,000 budget over twelve rounds; agent 2 is the saboteur. Three scenarios raise the difficulty. In *Simple* the saboteur flip-flops openly and then reforms after round 8; in *Subtle* it drifts in polite, hedged language while agent 1 is a loud-but-honest decoy; in *Inject* the subtle saboteur additionally embeds a prompt-injection clause that frames an honest agent. A frozen NLI head scores each utterance for entailment and contradiction of “the budget is fair”; a persistently incoherent agent crosses its summary bound ($L_j > U_j$); and a 6×6 trust matrix, trained by the Łukasiewicz necessity of coherence, drains that agent’s column, so the predicted saboteur is $\arg \min_j \overline{A_\theta[\cdot, j]}$. The main paper gives the loss and the headline detection numbers; the exact prompts, seeds, quantisation and training schedule are in the released code.

Motivation. The advantage over the LLM-judge comes from the front-end, not the modal layer: scoring each utterance against a fixed claim, rather than judging the transcript as a whole, is what lets the pipeline discount the decoy’s tone. Indeed the modal objective and a plain per-edge residual reach their optima together, so they detect identically and this experiment cannot separate them.

Proposition 2 *Let $c_j = \text{consistent}(j) \in [0, 1]$ and $A \in [0, 1]^{n \times n}$. In the hard limit $\tau \rightarrow 0$,*

$$\Box_i \text{coherent} = \min_j (1 - A_{ij} + c_j) = 1 \text{ for all } i \iff \sum_{i,j} \text{relu}(A_{ij} - c_j) = 0,$$

both being equivalent to $A_{ij} \leq c_j$ for all i, j .

Proof $\min_j (1 - A_{ij} + c_j) = 1$ holds iff $A_{ij} \leq c_j$ for every j (the term is clamped at 1); the residual is a sum of non-negative terms and vanishes under the same condition. ■

What the modal formulation adds here is therefore not accuracy but two things a transcript-level judge lacks: a per-cell audit trail, and an explicit belief-revision policy (below).

Detection. The main paper reports detections pooled over the five backbones. The per-backbone breakdown (Table S4) shows where the effect lives: the entire separation comes from the four 7–8B backbones, while *qwen2.5:14b* is at ceiling for *both* methods in every scenario. The supported claim is thus conditional and the pipeline’s advantage over a global LLM-judge appears at the 7–8B scale and closes at 14B, where the judge already solves the task, thus we leave it falsifiable at 30B+ rather than claim an unconditional win.

Table S4: Doxastic saboteur detection per backbone, MLNN (per-utterance NLI + learned trust matrix) vs. an LLM-judge reading the same transcript globally, correct out of three ollama seeds. The separation is concentrated on the 7–8B backbones, where qwen2.5:14b is at ceiling for both methods.

Backbone	Simple		Subtle		Inject	
	MLNN	judge	MLNN	judge	MLNN	judge
llama3	3	3	3	0	3	0
mistral	3	3	2	0	3	0
qwen2.5:7b	3	3	3	0	3	1
qwen2.5:14b	3	3	3	3	3	3
gemma:7b	3	3	1	0	3	1
Total	15	15	12	3	15	5

Scorer. The three Subtle misses are a limit of the frozen NLI scorer, which under-fires on the softest hedges, not of the modal layer. Passing a content-focused LLM scorer through the identical pipeline (per-utterance score, consistency summary, the same trust matrix) closes the gap to 15/15 on Subtle with no regression elsewhere. Scorer-varying ablations confirm that the valuation is the component that moves detection: it is an interface, and a stronger scorer improves it without touching the logic.

Belief revision. What is distinctive here is that whether the model forgives a reformed saboteur is an explicit setting, not a hidden property of the detector. In the Simple scenario the saboteur stops sabotaging after round 8, and whether its trust recovers depends on a single choice of how many past rounds the coherence score is computed over. The default uses all rounds, so the early contradictions always count: trust stays collapsed and the agent cannot regain it by cooperating later, a useful security property. A forgiving variant uses only the last four rounds, so once the agent has cooperated long enough the old contradictions fall outside the window and its trust returns. Both variants run in the same pipeline; only the length of the window changes.

Relation. The artefact is the trust matrix itself, read as which agent the others should stop believing. Figure S1 shows the seed-mean matrix per backbone. Where MLNN detects the saboteur cleanly (llama3, qwen2.5:7b and qwen2.5:14b, each 3/3), its column A_2 is drained to a mean in-trust of 0.03–0.15 against 0.8–0.9 for the honest agents; the two weaker backbones leave it only partly drained (mistral 0.62 at 2/3, gemma:7b 0.68 at 1/3, where a different agent is usually blamed). The figure is thus the visual counterpart of the MLNN column of Table S4, and every drained cell is traceable to the utterances that caused it (the per-round collapse is in Figure S2, by a median round 4 of 12). Read as a frame (Section A.5), the one axiom that clears its shape-matched null is a *symmetry deficit*: the saboteur’s incoming trust collapses while its outgoing trust does not, a directed-distrust signature rather than transitivity.

D.2. Turbofan Wear (C-MAPSS)

Overview. This is the deontic case: obligation and permission read off jet-engine sensors. The question it puts to the framework is not accuracy but applicability: whether a safety rule written as modal logic can be trained on real, high-dimensional data and then read back, per cycle, as an

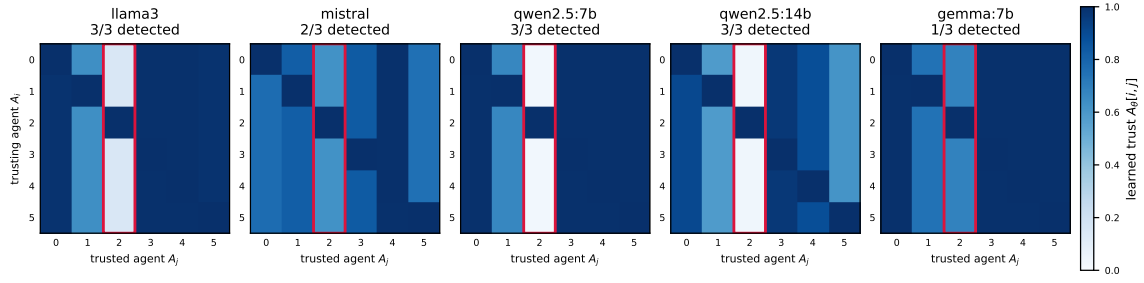


Figure S1: Learned trust matrices A_θ on the Subtle scenario, one panel per backbone, each the mean over the three seeds; agents are indexed 0–5 and the saboteur’s column (A_2) is boxed, with the MLNN detection rate in the title. Where detection succeeds the boxed column is drained (pale) while the honest columns stay trusted (dark); on **gemma:7b** (1/3) it stays trusted, because a different agent is usually blamed. This is the visual counterpart of the MLNN column of Table S4. All five matrices are the committed A_θ arrays; by Proposition 2 they show the artefact the modal layer produces, not that it detects better than a per-edge residual.

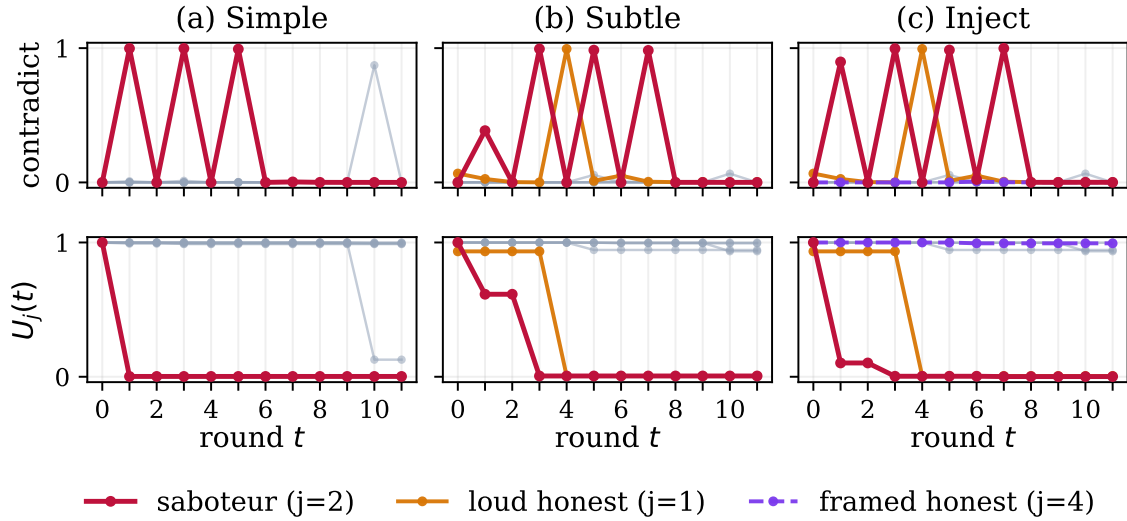


Figure S2: Per-round trust collapse (llama3) for **(a)** the loud flip-flopper (Simple), **(b)** polite drift with a loud-honest distractor (Subtle), and **(c)** drift plus an injection framing an honest agent (Inject). Top row: the per-round NLI contradiction score; bottom row: the running bound $U_j(t)$. The saboteur (**crimson**) collapses as soon as it acts (median round 4 of 12); the decoy (**amber**) keeps high trust because the front-end scores content, not tone; the framed agent (**violet**) stays flat, which is the injection immunity.

interpretable margin. It can. From the single rule that an engine must be safe and may be about to fail, but never both, the model learns a graded safety margin that tracks wear monotonically, extends to operating states it was given no label for, and reaches this at accuracy close to a standard classifier while remaining an object one can inspect.

Setup. NASA C-MAPSS FD002: 260 training engines (48,299 (engine, cycle) states) and 259 test engines (32,652 states), each a 24-dimensional reading (3 operating settings and 21 sensors). Two small heads read the sensor vector: an encoder $f_\theta (\mathbb{R}^{24} \rightarrow \mathbb{R}^{16})$, 2642 parameters) defining the accessibility relation through a metric kernel, and a valuation V_ϕ (1665 parameters) with $V_{\text{SAFE}} = 1 - V_\phi$. The two training signals are deliberately separated: labels reach only V_ϕ (a binary RUL < 30 target), while the relation is trained only by the deontic contradiction, with V detached,

$$\mathcal{L}_{\text{deon}} = \underbrace{(1 - U_{\Diamond(\text{FAIL-SOON})})^2}_{\text{RUL} < 30} + \underbrace{U_{\Diamond(\text{FAIL-SOON})}^2 + (1 - L_{\Box(\text{SAFE})})^2}_{\text{RUL} > 100}.$$

so the middle band ($30 \leq \text{RUL} \leq 100$) carries no loss term and is shaped only through the relation.

Bounds. Averaged by true state, the two bounds separate monotonically as an engine wears, and nothing reads RUL at inference:

state	worlds	mean $U_{\Diamond(\text{FAIL-SOON})}$	mean $L_{\Box(\text{SAFE})}$
safe (RUL > 100)	27,149	0.073	0.927
at-risk (30–100)	16,064	0.336	0.664
near-failure (RUL < 30)	5086	0.925	0.075

The at-risk row is the informative one: it has no term in the loss, yet its bounds fall in order between the safe and near-failure ends, so the modal read interpolates smoothly across a regime the model was never given a target for. That interpolation is evidence the relation generalises the safety structure rather than memorising two endpoints. We do not overstate the ends: with V_ϕ trained on the RUL < 30 label, the fail-soon bound there is close to a smoothed copy of that classifier. The label-free transition through the middle is the model’s own.

Interpretability. The margin is not bought with performance. On the held-out engines the fail-soon bound reaches AUROC 0.980, close to logistic regression on the raw sensors (0.988) and to the valuation head alone (0.990), from roughly four thousand parameters. And the learned space stays physically meaningful: its six operating regimes remain linearly separable (0.91, chance 0.17) and a probe recovers continuous RUL (Spearman 0.71) although only a binary label was supervised. This is preservation, not discovery (the raw sensors already carry the regimes at 0.93), but that is what interpretability needs: a purely logical objective leaves the engineering geometry intact and legible, so the learned state can be read after training rather than treated as a black box.

Frame structure. Read as a Kripke frame the relation is reflexive and symmetric, but this reflects the parameterisation, not learning: the inner-product kernel is symmetric by construction and its diagonal is largest by construction, while the implication axioms are near-vacuous on so sparse a relation. Under the matched null of Section A.5 no axiom clears its shape control, and a directed kernel trained identically does not change this. The audit’s role here is to name the relation; the result C-MAPSS carries is the interpretable deontic margin above.

D.3. Sentence Ordering (ROCStories)

Setup. ROCStories Spring-2016 + Winter-2017 Mostafazadeh et al. (2016): 98,155 deduplicated five-sentence stories (96,155 train after holding out 2,000 for in-domain validation); the 1,571 Story-Cloze-2018 Sharma et al. (2018) validation stories (rebuilt with the labelled correct ending) form an out-of-distribution test, disjoint from the training pool by story id and by full text. Each sentence is embedded by a frozen all-MiniLM-L6-v2 Reimers and Gurevych (2019) (384-d, ≈ 22.7 M frozen parameters); the only trainable object is the asymmetric rank-32 bilinear head $A_\theta(w_i, w_j) = \sigma(\alpha \langle P_{\text{src}} h_i, P_{\text{tgt}} h_j \rangle + \beta)$ with $P_{\text{src}}, P_{\text{tgt}} \in \mathbb{R}^{32 \times 384}$ (24,578 parameters). The score of a candidate order σ is the *Until* (adjacent- A_θ chain) U (is-last) evaluated at w_0 (the framework’s until operator) over all 120 permutations of S_5 ; the contradiction loss drives the true (identity) order above every scramble by a margin $m = 0.1$. AdamW, 50 epochs, 5 seeds; each checkpoint is selected by in-domain pairwise accuracy, so the Cloze columns are a genuine held-out estimate. A cosine-similarity ordering baseline (mean adjacent SBERT cosine, no learned head) sits at chance: pairwise 0.504, exact 0.008.

Ablation. *Exact* is the fraction of stories whose full order is recovered; *pair* is the Kendall-style fraction of sentence pairs placed in correct relative order (chance 0.5). *in* is held-out ROCStories, *cl* the out-of-distribution Story-Cloze-2018. All cells are 5-seed means (per-seed std ≤ 0.012); all configurations use the same contradiction loss.

Config	params	in pair	in exact	cl pair	cl exact
until_contradiction (ours)	24,578	0.788	0.225	0.765	0.235
no_margin ($m=0$)	24,578	0.737	0.148	0.708	0.142
symmetric ($P_{\text{src}}=P_{\text{tgt}}$)	12,290	0.581	0.080	0.566	0.074
+aux (+BCE, +hinge)	24,578	0.725	0.177	0.715	0.190
+multi-formula (+ G , + GF)	24,581	0.804	0.235	0.770	0.214

Stronger encoder The ablation above fixes the frozen encoder to all-MiniLM-L6-v2. A stronger frozen encoder (all-mpnet-base-v2, 768-d) with a rank-64 head ($P_{\text{src}}, P_{\text{tgt}} \in \mathbb{R}^{64 \times 768}$, 98,306 parameters) lifts MLNN to held-out pairwise 0.819/exact 0.327, on par with a prompted 7B LLM (qwen2.5-7b, 0.854/0.330) at $\sim 10^{-4}$ of its parameter count: the performance ceiling here is the frozen sentence embedding, not the modal logic. To isolate what the *structure* contributes, we add a supervised *pairwise head*, a 128-hidden-unit MLP on $[h_i; h_j; h_i - h_j]$ predicting $\Pr[w_i \prec w_j]$ ($\sim 3 \times$ MLNN’s parameters), trained with the same full order supervision on identical embeddings and decoded by the same 120-permutation argmax:

ordering model	encoder	cl pair	cl exact
cosine (no head)	MiniLM	0.504	0.008
	mpnet	0.495	0.014
pairwise head (supervised)	MiniLM	0.763	0.139
	mpnet	0.809	0.213
MLNN (until, ours)	MiniLM	0.763	0.236
	mpnet	0.819	0.327
qwen2.5-7b (prompted, zero-shot)	n/a	0.854	0.330

(Single seed 42; same 1,571 Cloze stories and per-story scramble for every row; per-seed std ≤ 0.012 as above. MLNN uses rank 32 (MiniLM) and rank 64 (mpnet).) Despite its extra parameters,

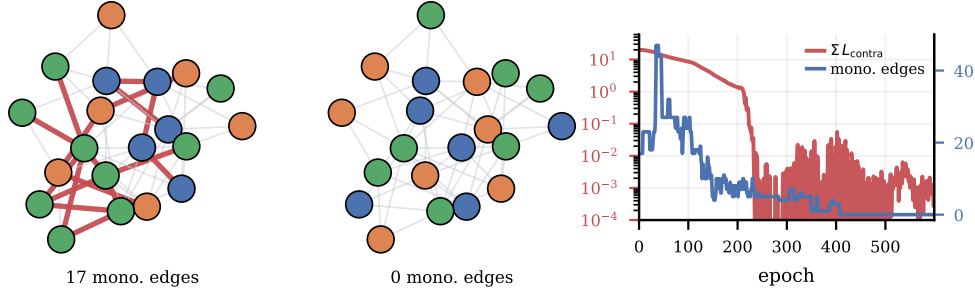


Figure S3: **Graph 3-colouring** ($L_{\text{task}}=0$, a 20-node instance). **(a)** epoch 1: 17 monochromatic edges (red) violate $p_c \rightarrow \neg \Diamond p_c$; **(b)** solved: 0 monochromatic edges; **(c)** both the per-node contradiction residual ΣL_{contra} (red, log scale) and the monochromatic-edge count (blue) drain to zero over the solve (ΣL_{contra} : $20.0 \rightarrow 1.4 \rightarrow 1.3 \times 10^{-3} \rightarrow 2.0 \times 10^{-4}$ and $17 \rightarrow 6 \rightarrow 0 \rightarrow 0$ monochromatic edges at epochs 1/200/400/600).

the head only *matches* MLNN on *local* precedence, *identical* on MiniLM at pairwise (0.763) and within 0.01 on mpnet, yet it recovers far fewer *full* orders (0.139 vs. 0.236 on MiniLM; 0.213 vs. 0.327 on mpnet). Its ten independent pairwise verdicts per story are mutually inconsistent, so no single permutation satisfies all of them; the *Until* recurrence enforces one globally consistent chain by construction. That global consistency, not local precedence (which the head learns equally well), is precisely what the modal structure adds.

Frame axioms. Of the four relations we audit, this is the one where a frame structure emerges rather than dissolving under the null. Trained only to tell coherent narratives from scrambled ones, the relation comes out directed: a sentence grants access to what follows it more than to what precedes it, so its symmetry falls below what a direction-blind relation would show ($B = 0.836$ against a shape-matched 0.864), and the asymmetry lines up with narrative time. That direction is what the task runs on, since forcing the relation to be symmetric collapses ordering accuracy (above), and it is the temporal counterpart of the doxastic trust matrix’s directed distrust; in the corrected audit it is the directed entry of Table S2. Transitivity, by contrast, is real but not distinctive: an order is transitive, yet so is a shape-matched control, so the finding is the direction, not the transitivity.

D.4. Graph Colouring

Setup. Planted-3-colourable graphs $G(n, p)$: a colouring is planted at random and edges are added only between differently-coloured nodes, so a proper 3-colouring is guaranteed to exist. Difficulty tiers (10 graphs each): easy ($n = 12$, $p = 0.30$), medium ($n = 20$, $p = 0.35$), hard ($n = 30$, $p = 0.40$). The MLNN solver realises the modal axiom $\bigwedge_c (p_c \rightarrow \neg \Diamond p_c)$ with $L_{\text{task}} = 0$ (Pure Satisfiability), free colour logits per node, Adam at learning rate 0.15, temperature annealed $0.5 \rightarrow 0.01$ over 2000 epochs, and 5 restart seeds; an instance counts as solved iff no edge is monochromatic. Every solver is wrapped in a hard per-call timeout; unavailable solvers are reported, not faked. The Pure-Satisfiability solve is illustrated in the body (the graph-colouring solve figure in the main paper); the inductive constraint-graph recovery (edge-recovery AUC = 1.000) is reported in the Graph Colouring experiment of the main paper.

Baseline matrix. Proper-colouring solve-rate (%) over the 10 graphs per tier:

Method	Family	Easy	Med.	Hard	Overall
Backtracking (DSATUR)	exact	100	100	100	100
OR-Tools CP-SAT	exact	100	100	100	100
Z3 (SMT)	exact	100	100	100	100
PycoSAT (SAT)	exact	100	100	100	100
python-constraint	exact	100	100	100	100
Greedy DSATUR	heuristic	100	100	100	100
Simulated annealing	metaheuristic	100	100	100	100
Min-conflicts	metaheuristic	100	100	100	100
MLNN (modal, ours)	differentiable	100	90	100	97
RRN-style GNN	differentiable	100	90	100	97
Semantic Loss	differentiable	100	90	70	87
Soft-nonmodal (ablation)	differentiable	20	90	70	60

The RRN-style GNN row is an unsupervised per-instance proxy of the supervised RRN [Palm et al. \(2018\)](#); SATNet [Wang et al. \(2019\)](#) is cited rather than run (see the complementarity paragraph below).

The differentiable methods are the relevant comparison class. MLNN and the RRN-style GNN tie at the top (97% each), the Semantic-Loss encoding [Xu et al. \(2018\)](#) follows (87%), and the non-modal ablation, the identical gradient solve with the $\Box/\Diamond$ operator replaced by the bare quadratic peer penalty $\sum_c p_c^\top A p_c$, trails at 60%. The modal operator clearly beats the bare penalty (97 vs 60), but, because a learned relational GNN ties it, *injecting relational structure* rather than the modal operator specifically is what drives the solve rate. Exact solvers and metaheuristics solve every instance, as expected of dedicated combinatorial methods on planted-colourable graphs; MLNN neither beats them nor is intended to. What MLNN adds beyond solving is the per-node contradiction residual L_{contra} , which drains to zero as the colouring becomes proper and localises the remaining conflicts at each step, and, in the inductive mode below, the recovered accessibility relation itself.

Positioning. Graph colouring is the constraint problem in which the modal logic is exact and the adjacency relation is the accessibility relation, and a proper colouring is the modal axiom $\bigwedge_c (p_c \rightarrow \neg \Diamond p_c)$ with no encoding overhead, which is why it is the natural stress test for a modal solver. The dedicated differentiable constraint solvers, SATNet [Wang et al. \(2019\)](#) (a learnable MaxSAT layer) and recurrent relational networks (RRN) [Palm et al. \(2018\)](#), are supervised: they learn the constraint rules from a corpus of solved instances (and SATNet requires a CUDA build), reaching $\approx 98\%$ and $\approx 96\%$ on their trained benchmarks respectively. They are strong solvers and we do not claim to beat them. MLNN is *complementary, not competitive*, on three axes a pure solver does not address: (i) it solves each instance directly from the given axioms with no training corpus (the RRN-style row above is an unsupervised per-instance proxy included for protocol parity; SATNet is supervised and CUDA-only, hence cited rather than run); (ii) in the inductive mode it *learns the accessibility relation itself* (edge-recovery AUC 1.000), which a fixed-rule solver never does; and (iii) it exposes the per-node L_{contra} residual as a step-by-step localisation of the still-violated constraints. The colouring contribution is thus not a solve-rate win but the demonstration that one modal mechanism both solves and recovers the constraint structure, with the relation inspectable throughout.

Recovery. Hiding the graph and supplying only 60 proper colourings of a 16-node instance, we optimise a symmetric, hollow $A_\theta = \sigma(M_\theta)$ to satisfy the colouring axiom across all supplied

colourings while maximising A_θ , so that the axiom, not a sparsity prior, fixes each entry. A pair of nodes that never shares a colour saturates towards 1 (it could be an edge), while a pair that shares a colour in any colouring is forced towards 0 by the axiom (it cannot be an edge). The recovered A_θ matches the held-out true adjacency at edge-recovery $\text{AUC} = 1.000$: the constraint graph, read off a learned accessibility relation from valid solutions alone. Identifiability is a property of the supplied colouring set, not of the axiom alone: the axiom forces A_{ij} towards 0 only for pairs that co-colour somewhere in the set, so a non-adjacent pair that shares no colour in any supplied colouring is indistinguishable from an edge and the maximisation saturates it to 1. With too few or insufficiently diverse colourings, several relations therefore fit the data; the 60 colourings used here are diverse enough that every non-edge co-colours at least once, which is what makes the recovery exact. Recovery is limited by colouring *supply*, not by size: at the body’s density ($p=0.25$) it is exact at $n=16$ ($\text{AUC} = 1.000$, 80 colourings) but falls to 0.78/0.74 at $n=24, 32$, where the denser graphs (54/96 edges) admit only 95/35 distinct colourings; on sparser graphs ($p=0.12$), where colourings stay plentiful, recovery is near-exact to 32-node hidden graphs ($\text{AUC} = 1.000$ at $n=24$, 0.982 at $n=32$, with 144/192 colourings). The fall-off is thus the colouring-supply limit just described (denser graphs admit fewer distinct colourings), not a modal one.

Scaling. The body tiers stop at $n=30$; pushing further surfaces one bug-fix and one modal-direction improvement, both now in the released solver. (i) The committed fixed learning rate 0.15 silently *diverges* past $n \approx 50$ (solve-rate collapses to 0% at $n=100$); a size-adaptive rate (0.04 for $n>30$) with more restarts restores it (100% at $n=50$ –100), as its loss landscape simply has more local minima at scale. (ii) A *focal* contradiction residual, $\sum \text{relu}(\text{viol}) + \text{relu}(\text{viol})^2$, concentrates the gradient on the binding conflicts the mean-field penalty dilutes, buying $\sim +10\text{pp}$ over the non-modal ablation at moderate-hard density. Plain $\Diamond$ -modal otherwise tracks the non-modal penalty at scale (the body’s 97 vs 60 gap is specific to the easy tiers), and a *hybrid* that adds a small non-modal term to the focal $\Diamond$ axiom keeps the inspectable modal structure while matching the penalty’s robustness (100% at $n=50$ –150 where pure focal-modal collapses; Figure S4). These are utility/robustness gains; consistent with the body, the modal contribution remains the inspectable and recoverable structure, not a solve-rate win over dedicated solvers.

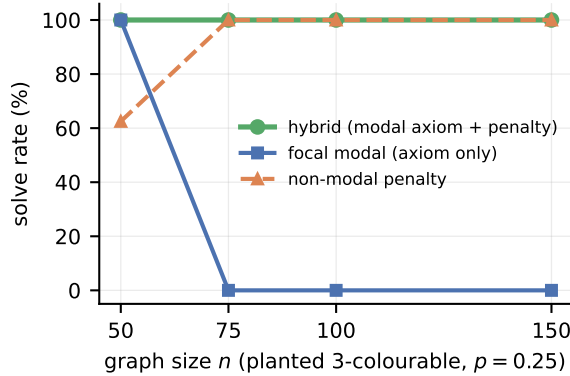


Figure S4: Graph-colouring solve rate versus size (planted 3-colourable, $p=0.25$, 8 instances, matched lr/restarts). Pure focal-modal collapses past $n=50$ and the non-modal penalty is robust, while the *hybrid* (focal $\Diamond$ axiom + a small non-modal term) holds 100% to $n=150$, keeping the inspectable modal axiom while gaining the penalty’s scale-robustness.

References

- Nasrin Mostafazadeh, Nathanael Chambers, Xiaodong He, Devi Parikh, Dhruv Batra, Lucy Vanderwende, Pushmeet Kohli, and James Allen. A corpus and cloze evaluation for deeper understanding of commonsense stories. In *Proceedings of NAACL-HLT*, pages 839–849, 2016.
- Rasmus Berg Palm, Ulrich Paquet, and Ole Winther. Recurrent relational networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018.
- Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of EMNLP-IJCNLP*, pages 3982–3992, 2019.
- Rishi Sharma, James Allen, Omid Bakhshandeh, and Nasrin Mostafazadeh. Tackling the story ending biases in the story cloze test. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 752–757, 2018.
- Po-Wei Wang, Priya L Donti, Bryan Wilder, and J Zico Kolter. SATNet: Bridging deep learning and logical reasoning using a differentiable satisfiability solver. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, pages 6545–6554, 2019.
- Jingyi Xu, Zilu Zhang, Tal Friedman, Yitao Liang, and Guy Van den Broeck. A semantic loss function for deep learning with symbolic knowledge. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, pages 5502–5511, 2018.

